Wire Pay Merchant & Connecting-Client Integration Specification

Item	Value
Version	1.1.0
Target system	Wire Pay (payment management dashboard) — commit `cf1c153` (`main`)
Date / revised	2026-09-06 (1.0.0) / 2026-09-07 (1.1.0; see the revision history at the end and `docs/spec/CHANGELOG.md`)
Language	English (Japanese edition: `docs/spec/ja/A-merchant-integration-spec.md`)
Terminology	Follows `docs/spec/en/glossary.md`

Intended readers

This document is for parties that call Wire Pay's API and dashboards to accept payments:

- developers and operations staff of merchants (e-commerce businesses)
- downstream PSPs and wallet operators that accept payments through Wire Pay (they hold a Wire Pay API key and act as a client of `/api/v1`)

Connection requirements for the parties Wire Pay entrusts processing to (Stripe, PayPay, acquirers, konbini collection agencies, virtual-account banks, upstream PSPs / wallets) are in the companion document Specification B (Payment Provider Connection Specification) `docs/spec/en/B-provider-connection-spec.md`.

How this document is written

- The body contains only facts verifiable from the implementation (Prisma schema, Route Handlers, the OpenAPI document, dictionaries, permission definitions, cron definitions, the admin manual).
- Unimplemented or planned functions are kept out of the body; they are listed at the end of each chapter under "Not implemented / planned" and in chapter 14.
- Statements that involve inference carry the marker "※要確認" (to be confirmed) and are listed with reasons in `docs/spec/ASSUMPTIONS.md`.
- Unless stated otherwise, dates are evaluated in Japan Standard Time (Asia/Tokyo); API timestamps are ISO 8601 / UTC. All amounts are whole yen (JPY has no minor unit).

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1. Overview and system structure

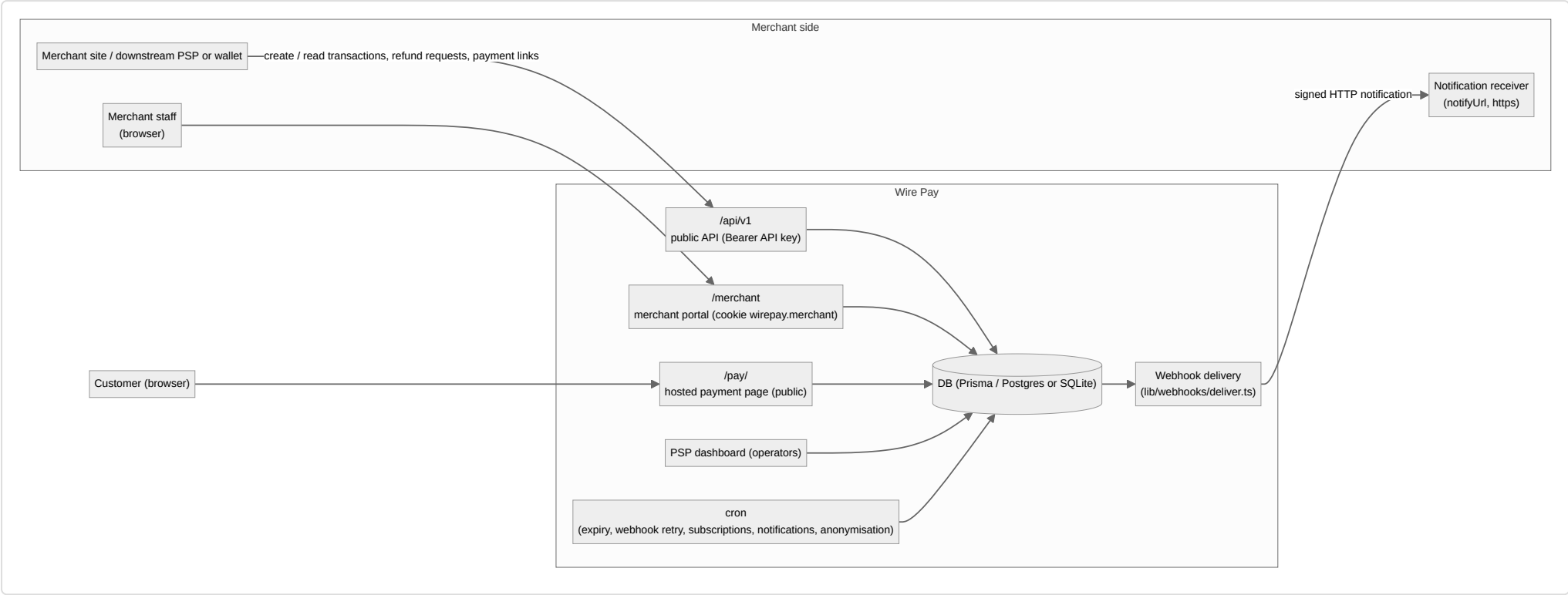
1.1 Wire Pay's role

Wire Pay consists of a payment management dashboard for the PSP (the operator dashboard), a public API for merchants (**`/api/v1`**), a merchant portal (**`/merchant`**) where merchants log in themselves, and a hosted payment page (**`/pay/<token>`**) for customers.

From the merchant's point of view Wire Pay does the following.

Role	Content	Chapter
Accepting transactions	Credit / QR / konbini / bank-transfer transactions are created by API or payment link	5, 7
Confirming payment	Deposit reconciliation for konbini and bank transfers, capture for cards, and the result delivered by webhook	6, 7
Refunds, voids, chargebacks	The PSP reviews the merchant's refund requests and reflects them in the transaction status	8
Settlement	Completed transactions minus fees, refunds and chargebacks are managed as settlement batches; a qualified invoice PDF is produced	9
Risk management	IP restrictions, transaction limits, payment-method suspension and the blacklist control what is accepted	10
Notifications	E-mails to customers, announcements and incident information to merchants	11

1.2 System structure



- The public API, merchant portal, payment page and PSP dashboard are one Next.js application (App Router) sharing one database and one set of business rules (`createTransaction` in `lib/payments/create.ts`, `applyPaymentAction` in `lib/payments/actions.ts`). A transaction created through any path goes through the same state machine, the same controls (limits, suspension, blacklist) and the same webhook delivery.
- The merchant portal session (cookie `wirepay.merchant`) is completely separate from the PSP dashboard session (chapter 3).
- External payment networks are not called (demo mode). Card / QR authorization and capture are processed immediately as database state transitions (Specification B, chapter 11).

1.3 Test and live environments

Wire Pay separates "test" and "live" on the same endpoints and the same database.

Aspect	Specification
How the environment is chosen	In the API it is the kind of API key (`wp_test_ak_…` = test, `wp_live_ak_…` = live). The request body cannot select the environment
Data separation	Transactions, refunds, settlement batches, payment links, subscriptions and webhook deliveries carry `livemode` (true = live). Data created with a test key is invisible to a live key (`GET /transactions/{id}` answers 404)
`livemode` of a transaction	The inverse of the provider's `testMode` is snapshotted when the transaction is created. Changing the provider's mode later never re-labels past transactions
Dashboards	The Test / Live switch in the header (cookie `wirepay.mode`, default test) selects the environment shown
Test cards	In test mode a card number reproduces a scenario (success / authorized only / failures). A test number sent to live answers 400 `test_card_in_live` (chapter 12)
Real charges	The target system runs in demo mode; no charge is ever sent to an external payment network in either environment (Specification B, chapter 11)

1.4 Supported payment methods

A merchant may use only the methods enabled by its contract (the merchant's "enabled methods", `enabledMethods`).

Method (**`method`** in the API)	Display name	Status right after creation	What completes it
`credit`	Credit card	`captured` (`autoCapture: true`, default) or `authorized` (`autoCapture: false`)	Completes at creation. An authorized transaction is completed from the PSP dashboard ("Complete")
`qr`	QR code payment	Same as `credit`	Same

Method (**`method`** in the API)	Display name	Status right after creation	What completes it
`convenience`	Konbini payment	`awaiting_payment` + 13-digit receipt number + payment deadline	The store's payment notification (the `deposit` action)
`bank`	Bank transfer	`awaiting_transfer` + 7- or 8-digit transfer ID + transfer deadline	The bank's deposit notification or the PSP's reconciliation (the `deposit` action)
`other`	Other	Cannot be requested through the public API (`API_METHODS` are the four above)	—

Per-method flows are in chapter 7, status transitions in chapter 2.

1.5 Connection paths covered

Path	Authentication	Purpose	Chapter
Public API `/api/v1/*`	`Authorization: Bearer wp_<test|live>_ak_…`	Create / read / list transactions, refund requests, payment links	5
Merchant API `/api/merchant-api/*`	Same API key	Connectivity check (`ping`), settlement batches, announcements	5.8
Webhooks (Wire Pay → merchant)	`X-Signature` (HMAC-SHA256)	Results of transactions, refunds, chargebacks, subscriptions	6
Merchant portal `/merchant/*`	Merchant ID + password (cookie `wirepay.merchant`)	Transactions, refunds, webhook log, settlements, API keys, method status, announcements, sandbox	3, 4
Hosted payment page `/pay/<token>`	None (public)	Customers pay through a payment-link URL	7.5
Receipt download `/api/pay/<token>/receipt`	Signed query (valid 30 days)	Customer downloads the receipt PDF	7.5

1.6 Not implemented / planned (chapter 1)

- There is no live connection to external payment networks. Card / QR complete immediately; konbini and bank deposits are simulated by the demo endpoints (chapter 12) or PSP dashboard actions.

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2. Terminology and the transaction lifecycle

2.1 Payment applications and transactions

In Wire Pay the record created when a customer picks a payment method on the merchant's site is a transaction, one row of the `Payment` table. Transactions before completion appear on the PSP dashboard as "Payment applications", completed / refunded / charged-back transactions as "Payments" — the same table filtered by status.

List	Statuses (**`lib/payments/transitions.ts`**)
Payment applications (`APPLICATION_STATUSES`)	`pending` , `awaiting_transfer` , `awaiting_payment` , `authorized` , `failed` , `expired` , `cancelled`
Payments (`SALES_STATUSES`)	`captured` , `partially_refunded` , `refunded` , `chargeback`
Open (dashboard KPI, `OPEN_STATUSES`)	`pending` , `awaiting_transfer` , `awaiting_payment` , `authorized`
Waiting for money (carry a deadline in `transferDueAt` , `WAITING_STATUSES`)	`awaiting_transfer` , `awaiting_payment`

The glossary (`glossary.md`) is authoritative for terminology. The most frequent terms of this chapter:

Term	Meaning
Transaction ID (`payment_id` / `id` in the API)	Wire Pay's unique transaction identifier (cuid)
Order number (`orderNumber`)	The merchant's order number (1–64 characters). Uniqueness is the merchant's responsibility
Authorization	Reserving the funds for a card / QR payment. Status `authorized`
Capture	Completing an authorized transaction. Status `captured`
Void (`cancel`)	Abandoning a transaction before completion. Status `cancelled` (UI label "Cancelled")
Refund	Returning money on a completed transaction. `partially_refunded` / `refunded`
Transfer ID (`transferId`)	The 7- or 8-digit number a bank-transfer customer prefixes to the payer name
Receipt number (`cvsReceiptNo`)	The 13-digit number a konbini customer quotes at the counter
Deposit reconciliation (`deposit`)	Matching a bank / konbini deposit to a transaction and completing it

2.2 Statuses

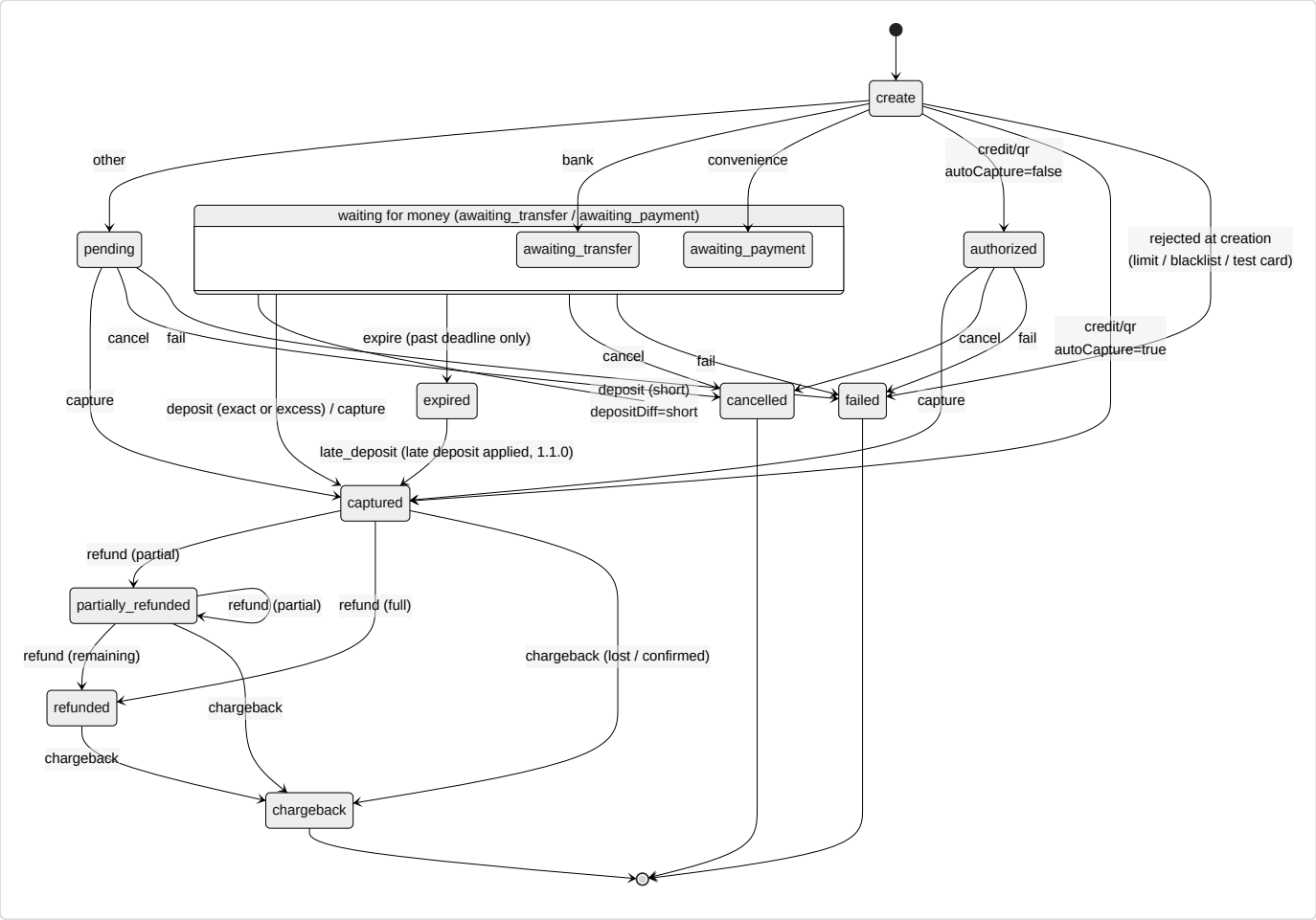
Status (DB value)	Label (ja / en)	Meaning
`pending`	申請中 / Pending	Just created, nothing confirmed (the English label changed from "Applied" to "Pending" in 1.1.0)
`awaiting_transfer`	振込待ち / Awaiting transfer	Bank transfer; transfer ID issued, waiting for the deposit

Status (DB value)	Label (ja / en)	Meaning
`awaiting_payment`	支払待ち / Awaiting payment	Konbini; receipt number issued, waiting for the counter payment (added in 1.1.0; `pending` was reused before)
`authorized`	与信済 / Authorized	Card / QR authorization held (`autoCapture: false` , or the "authorized only" test card)
`captured`	完了 / Completed	Payment settled; `capturedAt` stamped
`partially_refunded`	一部返金 / Partially refunded	Refunded total below the amount
`refunded`	返金済 / Refunded	Refunded total reached the amount
`cancelled`	キャンセル / Cancelled	Abandoned before completion (`cancelledAt`)
`failed`	失敗 / Failed	Rejected at creation (limit, blacklist, test card) or failed manually / by webhook. Reason in `failureReason`
`expired`	期限切れ / Expired	Transfer / payment deadline passed. A deposit that arrives afterwards is held as an unmatched deposit and can be applied by the PSP through `late_deposit` (7.3)
`chargeback`	チャージバック / Chargeback	A chargeback was lost or confirmed

Values of `failureReason` are listed in 5.3 ("Values of `failureReason` ").

2.3 State diagram

The transitions are defined once in `TRANSITIONS` (`lib/payments/transitions.ts`); the API, dashboards, webhooks and cron all derive from it. Updates are conditional on the current status being an allowed source, so concurrent requests can never apply twice (a conflict answers 409 `invalid_transition`).



Actions, sources and targets (`TRANSITIONS`):

Action	From	To (nominal)	Timestamp stamped	Who triggers it
`capture`	`pending` , `authorized` , `awaiting_transfer` , `awaiting_payment`	`captured`	`capturedAt`	PSP dashboard (permission `payment.state`), the hosted page's "Complete payment (demo)", provider webhook `payment.captured`
`cancel`	`pending` , `authorized` , `awaiting_transfer` , `awaiting_payment`	`cancelled`	`cancelledAt`	PSP dashboard, provider webhook `payment.cancelled`
`refund`	`captured` , `partially_refunded`	`refunded` (`partially_refunded` when partial)	`refundedAt` (full)	Approval of a refund request (chapter 8), provider webhook `payment.refunded`
`fail`	`pending` , `authorized` , `awaiting_transfer` , `awaiting_payment`	`failed`	—	PSP dashboard, provider webhook `payment.failed`
`expire`	`awaiting_transfer` , `awaiting_payment`	`expired`	—	The expiry cron (past deadline only; schedule in Specification B 8.4), PSP dashboard

Action	From	To (nominal)	Timestamp stamped	Who triggers it
`deposit`	`awaiting_transfer`, `awaiting_payment`	`captured` (stays put when short)	`capturedAt` (on completion)	Bank / konbini deposit notification (demo API), PSP reconciliation (permission `payment.deposit`), matching an unmatched deposit
`late_deposit`	`expired`	`captured` (completes even when short / excess; the difference is kept in `depositDiff`)	`capturedAt`	"Apply as late deposit" on the unmatched-deposit screen (permission `unmatched_deposit.match`, reason required; added in 1.1.0; never triggered by the API or webhooks)
`chargeback`	`captured`, `partially_refunded`, `refunded`	`chargeback`	—	A chargeback lost or confirmed (never a direct dashboard action)

2.4 Differences per payment method

Credit / QR (authorization and capture)

- With `autoCapture: true` (default) the transaction is `captured` at creation and `payment.completed` is delivered.
- With `autoCapture: false` it stops at `authorized`; the PSP dashboard's "Complete" moves it to `captured`. There is no merchant API to capture (chapter 14).
- An authorized transaction can be voided with "Cancel" (not a refund).

Bank transfer (transfer ID, deadline, short / excess deposits)

- A transfer ID of the merchant's configured length (7 or 8 digits, `transferIdDigits`) is issued at creation. Its first digit is never 0 and it is unique system-wide.
- Transfer deadline = creation time + the merchant's `transferDueDays` (default 7 days, 1–60).
- Deposits are evaluated cumulatively (`reconcileDeposit`): exact → `captured`; short → stays `awaiting_transfer` with `depositDiff = "short"` (the next deposit accumulates); excess → `captured` + `depositDiff = "excess"`. Short / excess cases are tracked on the PSP's "Short / excess deposits" screen: the excess is refunded through a refund request, the shortfall triggers an additional-deposit e-mail.
- Awaiting transfers past the deadline are expired by the cron `/api/cron/expire-transfers` (webhook `payment.expired`; the demo's `vercel.json` runs it once a day, hourly is recommended in production — Specification B 8.4).
- A deposit that arrives without a transfer ID is recorded as an "unmatched deposit"; PSP support attaches the transfer ID and the same reconciliation runs (`payment.completed` is delivered on completion).
- Deposits that arrive after expiry or cancellation, and deposits whose environment (test / live) does not match, are not reconciled but held as unmatched deposits (1.1.0). A late deposit applied by the PSP (reason required) moves the transaction to `captured` and delivers `payment.completed` and the completion e-mail as usual; a deposit on a cancelled transaction can only be dismissed as a return (7.3).

- A re-sent deposit notification (same `eventId`, or identical content) is never counted twice (1.1.0).

Konbini (receipt number, payment deadline)

- A 13-digit receipt number (first digit never 0, unique) and the store (`cvsBrand`: `seven` / `lawson` / `familymart` / `ministop` / `other`, default `seven`) are recorded at creation.
- Payment deadline = creation time + the merchant's `cvsDueDays` (default 3 days, 1–30), stored in the same column as the transfer deadline (`transferDueAt`).
- The initial status is **`awaiting_payment`** (added in 1.1.0; `pending` was reused before). The store's notification runs the same `deposit` action as a bank deposit and stamps `cvsPaidAt` on completion.
- Expiry, late deposits, environment mismatches and re-sent notifications are handled exactly as for bank transfers.

Common

- On completion (`capture`, or an exact / excess `deposit`) the customer receives a "payment completed" e-mail; on a completed refund a "refund completed" e-mail (only when the transaction has a customer e-mail, chapter 11).
- Every status change is recorded in `PaymentEvent` (status history) with its actor (`dashboard` / `api` / `webhook` / `bank` / `cvs` / `system` / `seed`). The merchant portal's transaction detail shows this history.

2.5 Not implemented / planned (chapter 2)

- There is no capture or cancel through the merchant API; completing or voiding an authorized transaction is a PSP dashboard action.

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3. Accounts and permissions

3.1 Issuing a merchant account

Merchant accounts are issued by Wire Pay administrators (PSP operators) on the PSP dashboard ("Merchants → + Add merchant"). Merchants cannot sign up themselves.

Item	Specification
Merchant ID (`merchantCode`)	Auto-numbered `MER-00001`; immutable
Merchant password	Generated at registration and stored AES-GCM encrypted. Only the admin role can reveal / reissue it on the PSP dashboard (reissuing invalidates the old password immediately)
Site name / kana (full-width katakana) / company / contact / e-mail / statement e-mail	Required
Notification URL (`notifyUrl`)	Webhook destination; https only
Notification parameters	Format / parameter set / identifier format (chapter 6)

Item	Specification
E-mail notification setting (`emailNotification`)	`per_payment` / `daily` / `none` (chapter 11)
Agent	Optional
Enabled methods (`enabledMethods`)	Any of `credit` / `qr` / `convenience` / `bank` / `other`. Methods not listed answer 400 `method_not_enabled` in the API
Transfer ID length (`transferIdDigits`)	7 or 8

Registration issues the test-environment signing secret (**`wp_test_ss_…`**) automatically and shows its plaintext once on the completion screen (1.1.0). API keys (`api_key`) and the live signing secret (`signing_secret`) are issued by the PSP per environment (test / live) on the merchant detail's API keys tab. The plaintext is shown once; the database keeps only a SHA-256 hash (signing secrets additionally keep a ciphertext for webhook signing). Webhooks are not delivered to an environment without a usable signing secret (6.4).

3.2 API keys and signing secrets

Kind	Format	Use
API key	`wp_test_ak_<32 chars>` / `wp_live_ak_<32 chars>`	Bearer authentication for `/api/v1/*` and `/api/merchant-api/*`
Signing secret	`wp_test_ss_<32 chars>` / `wp_live_ss_<32 chars>`	Shared key for verifying webhook `X-Signature`. Cannot authenticate API calls (401 `invalid_api_key`)

Key states (`apiKeyStatus`):

State	Condition	Authenticates
`active`	Neither revoked nor expiring	Yes
`expiring`	Old key after a rotation, `expiresAt` in the future	Yes
`expired`	`expiresAt` passed	No (401)
`revoked`	Revoked	No (401)

- Rotation issues a new key with the same mode, kind and label and gives the old key an `expiresAt` after a grace period (default `ROTATION_GRACE_DAYS` = 7 days; the PSP may choose 0–90). Test keys can be rotated by the merchant in the portal (grace fixed at 7 days). Live keys are issued, rotated and revoked only by the PSP; a merchant trying to rotate a live key gets 403 `live_key_rotation`.
- Demo cap: 20 keys per merchant (`DEMO_LIMITS.apiKeysPerMerchant`).
- `lastUsedAt` is updated on every successful authentication.

3.3 Merchant portal login

Item	Specification (**`lib/merchant-portal/session.ts`**)
URL	`/merchant/login`
Credentials	Merchant ID (case-insensitive, `mer-00001` accepted) + merchant password

Item	Specification (**`lib/merchant-portal/session.ts`**)
Session	Cookie `wirepay.merchant` (HMAC-signed, purpose `merchant`, 12 hours). Key derivation is separate from the PSP dashboard cookie; neither can be used for the other
Lockout	5 consecutive failures lock the account for 15 minutes (HTTP 423 `merchant_locked`, remaining minutes returned). The correct password also answers 423 while locked
Rate limit	The login API is limited to 10 requests / minute per IP (shared budget with the PSP login)
Audit	Success `merchant.login`, lockout `merchant.lockout`, logout `merchant.logout` (actor type "merchant")
After a password reissue	Only the new password works; existing sessions remain valid until they expire

After login the environment switch (Test / Live, cookie `wirepay.mode`) selects the environment shown.

3.4 What merchants can and cannot do

Merchants are limited to the following through the portal, the public API and the merchant API.

Operation	Path	Notes
Create / read / list transactions	`/api/v1/transactions`	Own merchant, same environment only
Request refunds	`/api/v1/refunds`, portal transaction detail	Approval / rejection is the PSP's (admin / accounting). Merchants cannot force a refund past the window
Create payment links	`/api/v1/payment-links`	Disabling is PSP-dashboard only (chapter 14)
Resend webhooks	Portal	Own deliveries only
Rotate test keys	Portal	Live keys excluded
Read settlement batches / lines, export the invoice PDF	Portal, `/api/merchant-api/settlements`	Invoice for confirmed batches and later only (the invoice number is assigned at confirmation)
Issue receipts (own completed / partially refunded transactions)	Portal transaction detail	Addressee and description; PDF (4.5, 1.1.0)
Read payment-method status and announcements	Portal, `/api/merchant-api/announcements`	
Read own customers	Portal	Personal data unmasked (own data)
Call the API from the sandbox	Portal	Test keys only

Operations merchants cannot perform (PSP permissions):

Operation	Performed by
Completing (capturing), failing, cancelling or expiring an authorized transaction	PSP (`payment.state` = admin)
Deposit reconciliation, unmatched deposits, short / excess follow-up	PSP (`payment.deposit` = admin / accounting and others)

Operation	Performed by
Approving / rejecting refunds, forcing refunds past the window	PSP (admin / accounting; forcing: admin)
Registering, disputing and resolving chargebacks	PSP (admin / accounting)
Changing contract / fee rates, limits, deadlines, settlement settings, notification settings, enabled methods	PSP
Issuing / rotating / revoking live API keys, issuing signing secrets	PSP (admin / developer)
Suspending / resuming payment methods, IP restrictions, blacklist	PSP
Generating, confirming and paying settlement batches	PSP (admin / accounting)
Issuing receipts from the PSP dashboard	PSP (admin / accounting / cs). Merchants issue from the portal's transaction detail; customers obtain theirs through the payment page
Matching unmatched deposits, applying late deposits, returns	PSP (admin / accounting)
Reading internal notes (transaction notes)	PSP only (never shown in the portal)

3.5 PSP roles (reference)

PSP dashboard roles are `admin` / `accounting` / `cs` / `developer` / `viewer`; the permission matrix is defined once in `PERMISSIONS` (`lib/role.ts`). The merchant-relevant subset (the full table is in the PSP manual's "Introduction"):

Merchant-relevant operation	admin	accounting	cs	developer	viewer
Create / update merchants	○		○		
Reveal / reissue the merchant password	○				
Update contract / fees, limits, deadlines, settlement settings	○	○			
Issue / rotate / revoke API keys and signing secrets	○			○	
Complete / fail / cancel / expire transactions	○				
Deposit reconciliation	○	○			
Match unmatched deposits (including late deposits)	○	○			
Approve / reject refunds	○	○			
Generate / confirm / pay settlement batches	○	○			
Resend webhooks (PSP side)	○			○	
Suspend payment methods globally / per merchant	○		○ (per merchant only)		
Change IP restriction rules and the allow-list mode	○			○	

The complete permission matrix is in Appendix E.

3.6 Not implemented / planned (chapter 3)

- The merchant portal has no password change / reissue, no multiple staff accounts and no two-factor authentication (PSP admin accounts have TOTP).
- Merchants cannot issue, rotate or revoke live keys (ask the PSP).

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4. Dashboard screens

This chapter describes the merchant portal (**`/merchant/*`**) and the hosted payment page (**`/pay/<token>`**) screen by screen. The PSP dashboard is covered only where it configures merchant behaviour (4.17). Labels are dictionary values (`lib/i18n/messages/ja.ts` / `en.ts`); the header's "EN / 日本語" switch toggles the language.

4.1 Common layout

Element	Content
Header	Wire Pay logo (→ `/merchant`), "Merchant portal", Test / Live switch (when logged in), "API docs" link (→ `/docs/api`), language switch, logout (`POST /api/merchant-portal/logout` → `/merchant/login`). On phones the switch and the language toggle move to a second row
Live-mode indication	In live mode a red line at the left of the header and a red stripe "Live mode — showing real data" across the top are shown, and every page title carries a badge with the current mode. Switching test → live shows a warning toast (1.1.0)
Sidebar	Site name and merchant ID, the menu below, "Which environment (test / live) to show; matches the API key's environment.". The link to the PSP dashboard is shown only in the demo (`DEMO_LOGIN_HINT` set) (1.1.0)
Guard	Every page except login requires a merchant session; otherwise redirect to `/merchant/login`

Menu	Path	Section
Home	`/merchant`	4.3
Transactions	`/merchant/transactions`	4.4 / 4.5
Customers	`/merchant/customers`	4.13 / 4.14
Refund requests	`/merchant/refunds`	4.6
Webhook log	`/merchant/webhooks`	4.7
Settlements	`/merchant/settlements`	4.8 / 4.9
API keys	`/merchant/api-keys`	4.10
Payment methods	`/merchant/methods`	4.11
Announcements	`/merchant/announcements`	4.12
Sandbox	`/merchant/sandbox`	4.15

Payment-method names come from `CATEGORIES` in `lib/providers/registry.ts` (Credit Card Payments / QR Code Payments / Konbini Payments / Bank Transfer Payments / Other Web Payment Services). Status labels

are in Appendix B.

4.2 Login (`/merchant/login`)

Item	Content
Purpose	Log in to the merchant portal with merchant ID + password
Inputs	Merchant ID (`MER-00001` format, case-insensitive), password
Action	"Log in" → `POST /api/merchant-portal/login { merchantCode, password }`; success redirects to `/merchant`
Errors	Wrong password 401; 5 consecutive failures 423 ("Too many failed logins; the account is locked. Try again in {minutes} minutes"). Section 3.3
Other	Already logged in → redirect to `/merchant`. With `DEMO_LOGIN_HINT` set, demo accounts and a link to the PSP dashboard are shown (never in production). Footer link to the API docs. The inputs are named `merchant-code` (`autocomplete="off"`) / `merchant-password` so browsers do not autofill the PSP operator's login ID (1.1.0)

4.3 Home (`/merchant`)

An overview of the merchant's account. Everything is read-only; each card links to its screen. The header environment switch applies.

Card	Aggregation	Link
Completed today	Count and amount of transactions in a sales status (completed / partially refunded / refunded / chargeback) whose `capturedAt` is today (JST)	Transactions
Awaiting payment / authorized	Count of `pending` / `awaiting_transfer` / `awaiting_payment` / `authorized`	Transactions
Refund requests under review	Count of refund requests with status `requested`	Refund requests
Failed / retrying webhooks	Count of deliveries with status `failed` / `abandoned` / `retrying`	Webhook log
Next payout	Earliest scheduled payout date and net payout among batches in `confirmed` / `scheduled`; "None scheduled" when none	That batch / Settlements

Below: "Recent transactions" (8 newest: order number, method, created, amount, status), "Announcements" (currently published: title, kind, published date) and the current e-mail notification setting (per payment / daily digest / none; ask the PSP to change it, 11.1) (1.1.0).

4.4 Transaction list (`/merchant/transactions`)

Item	Content
Purpose	Browse the merchant's payment applications and completed payments
Filter	Status (All + 11 statuses); URL `?status=<value>` . Environment from the header switch
Order	Completed transactions first (`order: "sales"`), 20 per page
Columns	Order (+ transaction ID), Customer (name + customer id), Method, Amount (with `−refunded` when any), Status (with an "Overdue" badge when past the deadline), Created
Actions	The order number opens the detail. No CSV export, no operator actions

4.5 Transaction detail (`/merchant/transactions/[id]`)

Other merchants' transactions answer 404. Internal notes (the PSP's transaction notes) are never shown.

Section	Fields
Heading	Transaction {order number}, environment badge (Test / Live), status, overdue badge
Transaction	Transaction ID, amount, refunded (if any), product, customer (name, id, e-mail), method, created, completed, failure reason (raw `failureReason` , if set)
Bank transfer (when a transfer ID exists)	Transfer ID, deadline, deposited (total). Rose border when overdue
Konbini (when a receipt number exists)	Store, receipt number (4-4-5 grouping), deadline, paid at ("Unpaid" until paid)
History	Status history (type → resulting status, time)
Refunds	Refund requests (reason code, time, amount, status)
Receipt (completed / partially refunded transactions)	Addressee (≤120) and description (≤120; blank = previous value → product name → "Goods") then "Issue" → `POST /api/merchant-portal/receipts/{paymentId}` `{ recipientName?, description? }` ; the PDF is `GET` on the same path. Number `RC-YYYYMM-00001` (same sequence as the PSP dashboard); reissues keep the number and count +1. `source = merchant` , audit `receipt.issue` (1.1.0)

Request refund (form on the right):

Item	Content
Shown when	Status `captured` / `partially_refunded` ; otherwise "Only completed transactions can be refunded."

Item	Content
Refund amount	1 … refundable (amount − refunded − pending requests); default is the whole refundable balance
Reason	Customer request / Defective goods / Duplicate payment / Other (default: customer). "Other" requires details
Reason (details)	≤500 characters
Submit	`POST /api/merchant-portal/refunds` `{ paymentId, amount, reasonCode, reason }` → 201 `{ id, status: "requested", amount }`. No confirmation dialog
Past the window	The submit button is disabled and "This transaction is past its refund window…" is shown. Merchants cannot force it (contact PSP support)
Success	"Refund requested. The decision is also delivered by webhook (refund.completed)."
Errors	409 (not completed), 400 `refund_exceeds`, 409 `refund_window_expired`, etc. (5.6)

4.6 Refund requests (**`/merchant/refunds`**)

Item	Content
Purpose	Review the merchant's refund requests and their outcomes
Filters	None (environment switch only); 20 per page
Columns	Requested, Order (→ transaction detail), Amount, Reason (code + details; the decision note in red when rejected), Status (Requested / Approved / Rejected / Processing / Completed / Failed), Completed
Actions	None (requests are made from the transaction detail)

4.7 Webhook log (**`/merchant/webhooks`**)

Item	Content
Purpose	Inspect deliveries to the notification URL and resend manually
Data	Latest 100 rows of the selected environment; no filters or paging
Columns	Sent, Event, URL (method + URL), Status (Pending / Success / Failed / Retrying / Abandoned), Attempts, Response (HTTP status)
Action	"Resend" → `POST /api/merchant-portal/webhooks/{id}/resend` (no body). No confirmation dialog. The signature (**`X-Timestamp`** / **`X-Signature`**) is recomputed. Result: "Re-sent (response {status})." or "Re-sent but failed: {error}"
Audit	`webhook.resend` (actor: merchant)

Delivery details (request headers / body, attempt history) are not in the portal; they are on the PSP dashboard's webhook log detail.

4.8 Settlements (`/merchant/settlements`)

Item	Content
Purpose	Review settlement batches (sales minus fees, refunds and chargebacks) and payout dates
Data	Latest 100 of the selected environment
Columns	Period (→ detail), Invoice number (once confirmed), Gross sales, Net payout (red when negative), Status (Collecting / Confirmed / Scheduled / Paid / Held), Payout date (paid date once paid)

4.9 Settlement detail (`/merchant/settlements/[id]`)

Section	Fields
Summary	Invoice number (`INV-YYYYMM-<merchant number>-<2-digit sequence>`, assigned at confirmation; blank while collecting), gross sales, fee total, refund total, chargeback total, settlement fee (transfer fee), net payout, payout date, paid at
Lines	Occurred, kind (payment / refund / chargeback), order (→ transaction detail), method, amount (red when negative), fee
Invoice PDF	While collecting: "Available after the batch is confirmed." Otherwise `GET /api/merchant-portal/settlements/{id}/invoice` opens in a new tab (audit `settlement.invoice_export`)

Formulas are in chapter 9.

4.10 API keys (`/merchant/api-keys`)

Item	Content
Purpose	Review issued API keys / signing secrets and self-rotate test keys
Columns	Key (first 16 characters + ⋯, label), Kind (API key / Signing secret), Environment (Test / Live), Status (Active / Expiring / Expired / Revoked), Issued, Expires, Last used
Action	"Rotate" on active / expiring test keys. Confirmation "Replace test key {prefix} with a new one. The old key keeps working for 7 days." → `POST /api/merchant-portal/api-keys/{keyId}/rotate` (no body) → 201 `{ key, plaintext }`
Result	"New key issued. It cannot be shown again after you leave this page." + plaintext + copy button. The plaintext is shown once only
Live keys	"Live keys are read-only here."; rotation answers 403 `live_key_rotation`
Empty	"No API keys yet. Ask Wire Pay to issue one."

4.11 Payment methods (`/merchant/methods`)

A read-only screen that always shows the four rows credit / QR / konbini / bank transfer ("other" is excluded). Evaluation order:

1. Not in the contract's enabled methods → Not in contract

2. Suspended globally (Wire Pay stopped it for all merchants) → Suspended (all merchants); suspended for this merchant → Suspended (your account); with the reason and "Resumes {date-time}"
3. Otherwise → Available

Creating a transaction with a suspended method answers 409 `method_suspended` in the API, and the method disappears from the payment page (10.3).

4.12 Announcements (`/merchant/announcements`)

Cards for announcements that are published (status "published" and within the window) and target all merchants or include this merchant: kind badge (Notice / Maintenance / Incident), title, published time, body and, for incidents, the timeline (occurred → investigating → resolved). No actions.

4.13 Customers (`/merchant/customers`)

Item	Content
Purpose	The merchant's customer master (one row per customer id used in transactions) with stored cards and transaction history
Display	Own data, therefore personal data is not masked (the PSP dashboard masks it)
Search	"Search (customer id / name / e-mail)"; Enter applies `?q=` (≤100 characters). 20 per page
Columns	Customer (name or customer id → detail; "Anonymised" badge), E-mail, Transactions, Cards (stored tokens), Last transaction
Actions	None (anonymisation is PSP admin only)

4.14 Customer detail (`/merchant/customers/[id]`)

Section	Fields
Heading	Name (or customer id), customer id. When anonymised: "This customer's personal data was anonymised on {date}…"
Profile	E-mail, phone, first transaction
Stored cards	Brand, •••• last 4, expiry, last used (or created). Card numbers are never stored
Transactions	Order (→ transaction detail), created, amount, status

4.15 Sandbox (`/merchant/sandbox`)

A console that calls the real public API (`/api/v1`) from the browser with a test key. It is not a separate mock environment: requests that work here work unchanged with a live key.

Item	Content
API key	`wp_test_ak_…` only (`wp_live_` is refused client-side: "Only test keys can be used in the sandbox."). Stored only in the browser's sessionStorage (`wirepay.sandbox.key`)

Item	Content
Endpoint	One of the five endpoints of 5.2; path parameters / query string / JSON body ("Load example" inserts the sample)
Send	Direct browser `fetch("/api/v1/...", { Authorization: Bearer <key> })`. Shows the response (`HTTP {status} · {ms} ms` and JSON), an equivalent curl command and the test card list
Caution	Transactions created here are real test data shown on both dashboards, and webhooks are really delivered. The 60 / minute rate limit applies

4.16 Hosted payment page (`/pay/<token>`)

A public page for customers (no login, no dashboard chrome): the URL of a payment link (5.7 / 7.5).

Item	Content
Display	Merchant site name, product, description, amount. Test links show "Test mode — nothing is actually charged"
Unavailable	If disabled / expired / exhausted, only the reason is shown and there is no form (7.5)
Methods	The merchant's enabled methods ∩ not suspended ∩ an active provider exists in the same environment. None → "No payment method is available right now."
Inputs	Method (radio), store when konbini is chosen, name (required, ≤120), e-mail (optional, ≤200). No card details are collected
Submit	"Pay {amount}" → `POST /api/pay/<token>` → 201. Bank transfer shows transfer ID, amount and deadline (no button; the deposit is reconciled by the PSP). Konbini shows store, receipt number, amount, deadline and "Mark as paid at the store (demo)". Credit / QR / other show "Complete payment (demo)"
Completed	"Payment completed — Reference {order number}" and "Download receipt (PDF)" (signed URL valid 30 days)

4.17 PSP dashboard settings that affect merchants (reference)

Merchants cannot change these themselves; ask the PSP.

PSP screen (merchant detail tab)	Settings	Chapter
Basic	Site / company / contact, agent, enabled methods, transfer ID length, contacts	3.1
Contract & fees	Contract plan (Standard / Premium / Custom) and per-method rate / fixed fee, effective date, transfer deadline (days), refund window (days), konbini payment deadline (days)	7, 8, 9
Limits	Per-method per-transaction / daily / monthly caps and usage	10.2
API keys	Issue / rotate / revoke API keys and signing secrets	3.2

PSP screen (merchant detail tab)	Settings	Chapter
Notifications	Notification URL, format, parameter set, identifier, sample	6
Settlement	Cycle, payout term, bank account, settlement fee, registration number and address for invoices	9
Payment methods	Per-merchant suspension	10.3
Webhook log	Recent deliveries to this merchant (details, resend)	6
IP restrictions (`/ip-restrictions`)	Allow / block rules for the merchant API and the allow-list mode (target "merchant API")	10.1
Deposit notification log (`/deposit-notifications`) / Unmatched deposits (`/unmatched-deposits`)	Every bank / konbini notification, and held deposits: match, apply as late deposit, return	7.3

4.18 Not implemented / planned (chapter 4)

- The portal has no CSV export, no detailed transaction filters (method, period, order search) and no webhook delivery detail (headers / body).
- Merchants cannot change notification settings, enabled methods, deadlines or their password themselves.

---

5. API specification

This chapter is derived from the OpenAPI 3.0.3 document returned by `GET /api/v1/openapi.json` (generated by `lib/public-api/openapi.ts` from the Zod schemas in `lib/public-api/schemas.ts`). Endpoint additions and changes are always reflected there; where this text and the OpenAPI document differ, the OpenAPI document prevails.

5.1 Common rules

Item	Specification
Base URL	`https://<host>/api/v1` (`servers[0].url`)
Format	JSON requests and responses (`Content-Type: application/json`); amounts are whole yen
Authentication	`Authorization: Bearer <API key>` . Key format `wp_(test|live)_(ak|ss)_[A-Za-z0-9]{32}` . Anything other than `ak` (a signing secret) answers 401 `invalid_api_key`

Item	Specification
Environment	The key's `test` / `live` is the data environment; the endpoints are shared
IP restriction	When the merchant has block rules for target "merchant API", a matching caller receives 403 `ip_blocked`. When the PSP has switched the allow-list mode (merchant API) on, a merchant with at least one allow rule also gets 403 `ip_blocked` (`details.reason = "not_in_allowlist"`) from callers matching none of them; merchants without allow rules are unaffected (1.1.0, 10.1). The caller IP is the first hop of `X-Forwarded-For`, else `X-Real-IP`
Rate limit	60 requests per minute per merchant × environment (fixed UNIX-minute windows counted in the database; calls to `/api/merchant-api/*` consume the same budget, 1.1.0). Over the limit: 429 `rate_limited` + `Retry-After` (seconds) + `X-RateLimit-Limit` + `X-RateLimit-Remaining: 0`. Successful responses also carry `X-RateLimit-Limit` / `X-RateLimit-Remaining`. The counter is consumed after authentication and before business processing, so requests that fail with 4xx count too
Idempotency	`POST /transactions` accepts an `Idempotency-Key` header (or `idempotencyKey` in the body, 8–128 characters). A retry with the same key returns the same transaction (HTTP 201, identical shape) and never creates a duplicate. Keys are unique system-wide (`Payment.idempotencyKey @unique`)
Language	The error `message` follows the `x-lang` header (`ja` / `en`), else `Accept-Language`, default `ja`
CSRF / session	`/api/v1/*` is exempt from the middleware's session and CSRF checks (Bearer only)
Audit	Creations and refund requests made through the API are logged with actor type merchant and the merchant ID. The status-history actor is `api`

Response envelope

```
{ "ok": true, "data": { ... } }
{ "ok": false, "error": { "code": "invalid_input", "message": "Invalid input", "details": { "fieldErrors": { "amoun
t": "too_small" } } } }
```

`details` depends on the code. `invalid_input` carries `fieldErrors` (field → reason code); Zod validation errors carry Zod's `flatten()` output (`fieldErrors` / `formErrors`).

Retry guidance

Response	Recommended action
429 `rate_limited`	Retry after `Retry-After` seconds
5xx / network failure	Retry with the same **`Idempotency-Key`** (prevents duplicates)
4xx (other than 429)	Fix the request before retrying

5.2 Endpoints

Operation	Method / path	Success	Summary
`createTransaction`	`POST /transactions`	201	Create a transaction
`listTransactions`	`GET /transactions`	200	The merchant's transactions of the key's environment (newest first, cursor paging)
`getTransaction`	`GET /transactions/{id}`	200	One transaction
`createRefund`	`POST /refunds`	201	Request a refund
`createPaymentLink`	`POST /payment-links`	201	Create a payment link
—	`GET /openapi.json`	200	The OpenAPI document (no authentication)

5.3 POST /transactions — create a transaction

Starts a payment. Merchant and environment come from the API key; enabled methods, suspensions, limits and the blacklist apply.

Request body (**`createTransactionApiSchema`**)

Field	Type	Required	Constraints	Description
`method`	string	○	`credit` / `qr` / `convenience` / `bank`	Payment method
`orderNumber`	string	○	1–64 chars	Your order number (shown on the dashboards and in webhooks)
`amount`	integer	○	1–99,999,999	Amount in yen
`productName`	string | null		≤200	Product name
`customerName`	string | null		≤120	Customer name
`customerId`	string | null		≤64	Your customer id; the key of the customer master (10.6)
`customerEmail`	string | null		email, ≤200	Receives transfer / konbini instructions and completion mails
`description`	string | null		≤2000	Free text
`autoCapture`	boolean		default `true`	`credit` / `qr`: `true` completes immediately, `false` leaves it authorized
`card`	object			Card details (below); meaningful only for `credit`
`card.number`	string	○ (when `card` given)	13–19 digits, Luhn-checked	Card number. Never stored (only brand, last 4 and the fraud fingerprint)

Field	Type	Required	Constraints	Description
`card.expMonth`	integer	○	1–12	Expiry month
`card.expYear`	integer	○	2024–2099	Expiry year (4 digits)
`card.cvc`	string		3–4 digits	Security code; neither validated nor stored
`cvsBrand`	string		`seven` / `lawson` / `familymart` / `ministop` / `other`	`convenience`: the store chain; default `seven`
`idempotencyKey`	string		8–128 chars	Alternative to the `Idempotency-Key` header

Header: `Idempotency-Key` (optional, 8–128 chars). When both header and body are present, the header wins.

Processing order

1. API-key authentication → rate-limit consumption.
2. If the merchant's `enabledMethods` does not contain `method` → 400 `method_not_enabled`.
3. `method = credit` with `card`: a Luhn failure is 400 `invalid_input` (`fieldErrors["card.number"] = "invalid"`). A test card number (chapter 12) is 400 **`test_card_in_live`** in live; in test it selects the scenario (failure scenarios record `failed` + `failureReason`; "authorized only" forces `autoCapture` to `false`). The brand is derived from the leading digits (4 → visa, 51–55 / 22–27 → mastercard, 35 → jcb, 34 / 37 → amex, 30 / 36 / 38 → diners, otherwise visa).
4. The provider is the oldest (`createdAt` ascending) row with `status = active`, `category = method` and the key's environment (`testMode = !livemode`). None → 400 `provider_not_found`.
5. `createTransaction` (shared logic):
    1. If the `Idempotency-Key` already exists, return the existing transaction (`replayed`).
    2. Demo cap (1,000 transactions, `DEMO_LIMITS.payments`) → 403 `demo_limit`.
    3. Payment-method suspension (global → per merchant) → 409 `method_suspended` (`details: { scope: "global" | "merchant", method }`).
    4. Blacklist (card fingerprint, source IP, e-mail): a hit records the transaction as **`failed`** (`failureReason = "blocked:<card|ip|email>"`).
    5. Limits (only when not blacklisted): per transaction → daily → monthly; exceeding records **`failed`** (`failureReason = "limit_exceeded:<per_transaction|daily|monthly>"`) and delivers the webhook `limit.exceeded`.
    6. Customer master upsert (when `customerId` is given).
    7. Row creation. `bank` gets a transfer ID + deadline, `convenience` a receipt number + deadline + store. A uniqueness collision of the transfer ID / receipt number is redrawn once.
    8. A transaction created `captured` delivers `payment.completed` and the customer's "payment completed" e-mail (1.1.0; not for subscription charges); `bank` sends the "transfer instructions" e-mail, `convenience` the "konbini payment instructions" e-mail. With the merchant's e-mail setting "per payment" the merchant contact is notified too (11.1).
    9. A `failed` transaction feeds repeated-failure detection (blacklist candidates).

Limit hits, blacklist hits and test-card failures all return HTTP 201 with **`status: "failed"`** — not an error response. Check `failureReason`.

Response (`data`: Transaction)

Field	Type	Description
`object`	`"transaction"`	Constant
`id`	string	Transaction ID (`payment_id` in webhooks)
`orderNumber`	string	Your order number
`status`	string	Statuses of 2.2
`amount`	integer	Amount
`amountRefunded`	integer	Refunded total
`currency`	`"JPY"`	Constant
`method`	string	`credit` / `qr` / `convenience` / `bank` / `other`
`mode`	`"test"` | `"live"`	Environment
`productName`	string | null	
`customer`	object	`{ id, name, email }` (each string | null)
`description`	string | null	
`failureReason`	string | null	Table below
`bankTransfer`	object | null	Bank transfer only: `{ transferId, dueAt (ISO/UTC | null), depositAmount (cumulative) }`
`konbini`	object | null	Konbini only: `{ brand, receiptNo (13 digits, no hyphens), dueAt, paidAt }`
`card`	object | null	When card details were sent: `{ last4 }`
`createdAt` / `capturedAt` / `cancelledAt` / `refundedAt`	string | null	ISO 8601 / UTC

Values of `failureReason`:

Value	Meaning
`limit_exceeded:per_transaction` / `limit_exceeded:daily` / `limit_exceeded:monthly`	Merchant limit exceeded
`blocked:card` / `blocked:ip` / `blocked:email`	Blacklist hit
`card_declined` / `insufficient_funds` / `expired_card` / `incorrect_cvc` / `processing_error`	Test-card failure scenarios (test environment)
`charge_declined`	Demo subscription failure (7.6)
null	No recorded reason (including manual / webhook `fail`)

Example

```
POST /api/v1/transactions
Authorization: Bearer wp_test_ak_…
Content-Type: application/json
Idempotency-Key: order-2026-00042

{ "method": "credit", "orderNumber": "ORD-2026-00042", "amount": 4980,
  "customerId": "u_10021", "customerEmail": "taro@example.com",
  "card": { "number": "4242424242424242", "expMonth": 12, "expYear": 2028 } }
```

```
{ "ok": true, "data": { "object": "transaction", "id": "clx1a2b3c4d5e6f7g8h9i0j1k", "orderNumber": "ORD-2026-0004
2",
  "status": "captured", "amount": 4980, "amountRefunded": 0, "currency": "JPY", "method": "credit", "mode": "test",
  "productName": null, "customer": { "id": "u_10021", "name": null, "email": "taro@example.com" }, "description": n
ull,
  "failureReason": null, "bankTransfer": null, "konbini": null, "card": { "last4": "4242" },
  "createdAt": "2026-09-01T03:00:00.000Z", "capturedAt": "2026-09-01T03:00:00.000Z", "cancelledAt": null, "refunded
At": null } }
```

Bank transfer (excerpt): `"status": "awaiting_transfer", "bankTransfer": { "transferId": "4821390", "dueAt": "2026-09-08T03:00:00.000Z", "depositAmount": 0 }` Konbini (excerpt): `"status": "awaiting_payment", "konbini": { "brand": "lawson", "receiptNo": "4102993827716", "dueAt": "2026-09-04T03:00:00.000Z", "paidAt": null }`

Errors

HTTP	code	Condition
400	`invalid_input`	Zod validation failure, Luhn failure, invalid `merchantId` (internal)
400	`invalid_json`	Body is not JSON
400	`method_not_enabled`	Method not in the merchant's contract
400	`test_card_in_live`	Test card number in the live environment
400	`provider_not_found`	No active provider in the environment
400	`provider_disabled`	Selected provider disabled (normally unreachable)
403	`demo_limit`	Demo cap
409	`method_suspended`	Payment method suspended
409	`conflict`	Transfer ID / receipt number redraw also collided

5.4 GET /transactions — list transactions

Returns the merchant's transactions of the key's environment, newest `createdAt` first (`id` descending on ties).

Query	Type	Constraints	Description
`status`	string	≤32	Exact status filter (the 11 values of 2.2, including `awaiting_payment`)
`method`	string	`credit` / `qr` / `convenience` / `bank`	Method filter
`from`	string	`YYYY-MM-DD`	Created on or after (JST midnight)
`to`	string	`YYYY-MM-DD`	Created on or before (JST, inclusive)
`limit`	integer	1–100, default 20	Page size
`startingAfter`	string	≤64	Cursor: the `id` of the previous page's last transaction

Response: `{ "data": [Transaction, …], "hasMore": boolean }`. When `hasMore` is `true`, pass the last element's `id` as `startingAfter`.

5.5 GET /transactions/{id} — retrieve a transaction

Returns one transaction. Another merchant's or environment's transaction and an unknown id all answer 404 **`not_found`** (existence is not disclosed).

5.6 POST /refunds — request a refund

Requests a refund of a completed transaction. The PSP approves or rejects it; `refund.completed` is delivered on completion. Business rules are in chapter 8.

Field	Type	Required	Constraints	Description
`transactionId`	string	○	1–64	Transaction ID
`amount`	integer		1–99,999,999	Amount; omitted = the whole refundable balance
`reasonCode`	string		`customer` / `defect` / `duplicate` / `other`, default `customer`	Reason code
`reason`	string		≤500	Free text; required for `reasonCode = other`

Response (`data`: Refund):

Field	Description
`object`	`"refund"`
`id`	Refund request ID
`transactionId`	Transaction ID
`amount`	Amount
`kind`	`full` (≥ transaction amount) / `partial`

Field	Description
`reasonCode` , `reason`	As requested
`status`	`requested` right after the request (Appendix B)
`createdAt` , `completedAt`	ISO 8601 / UTC

Errors:

HTTP	code	Condition
404	`not_found`	Other merchant / environment / unknown transaction
409	`invalid_transition`	Transaction not `captured` / `partially_refunded`
400	`invalid_input`	`other` without `reason`
400	`refund_exceeds`	`amount` > refundable balance (amount − refunded − pending requests). No `details` ; the message states the balance
409	`refund_window_expired`	Past the refund window (`capturedAt` + the merchant's `refundWindowDays`). Cannot be forced through the API

5.7 POST /payment-links — create a payment link

Issues a hosted payment page URL (7.5). The link's environment is the key's environment.

Field	Type	Required	Constraints	Description
`productName`	string	○	1–120 (trimmed)	Shown on the page
`amount`	integer	○	1–9,999,999	Amount in yen. Lower cap than the transaction API
`description`	string		≤500	Shown on the page
`expiresAt`	string		`YYYY-MM-DD`	Last valid day (JST, whole day)
`maxUses`	integer		1–100,000	Maximum number of checkouts (transaction creations); omitted = unlimited

Response (`data` : PaymentLink):

Field	Description
`object`	`"payment_link"`
`id`	Link ID
`url`	`https://<host>/pay/<token>`
`token`	24 characters (lower-case letters + digits)
`productName` , `amount` , `description`	Settings
`expiresAt`	Midnight JST of the day after the given date (ISO / UTC) or null

Field	Description
`maxUses` , `usedCount`	Cap and uses
`status`	`active` / `disabled`
`mode`	`test` / `live`
`createdAt`	ISO 8601 / UTC

5.8 Merchant API (**`/api/merchant-api/*`**)

Auxiliary endpoints authenticated with the same API key. IP restrictions (including the allow-list mode) and key states behave as in the public API, and the rate limit uses the same bucket as the public API (merchant × environment × minute, 60 per minute together with `/api/v1` ; 429 `rate_limited` + `Retry-After` when exceeded, `X-RateLimit-Limit` / `X-RateLimit-Remaining` on responses) (1.1.0; previously unlimited). The middleware treats `/api/merchant-api/` as a self-authenticating path, so no per-IP rate limit, CSRF or session check applies.

Endpoint	Response **`data`**
`GET / POST /api/merchant-api/ping`	`{ merchantCode, siteName, mode, kind, keyPrefix, status }` ; connectivity / key check
`GET /api/merchant-api/settlements`	`{ merchantCode, mode, settlements: [ { id, periodFrom, periodTo, status, invoiceNo, grossAmount, feeAmount, refundAmount, chargebackAmount, settlementFee, netAmount, scheduledAt, paidAt, holdReason, createdAt } ] }` (the key's environment, latest 100; `invoiceNo` is assigned at confirmation and null while collecting, 1.1.0)
`GET /api/merchant-api/settlements?id=<batchId>`	`{ batch, lines: [ { kind, method, description, amount, fee, occurredAt } ] }` ; other merchants / environments answer 404
`GET /api/merchant-api/announcements`	`{ merchantCode, announcements: [ { id, kind, title, body, publishFrom, publishUntil, updates } ] }` ; published announcements targeting everyone or this merchant

5.9 Extra information in the OpenAPI document

Besides the standard `paths` / `components` , `GET /api/v1/openapi.json` carries these extension fields.

Field	Content
`x-error-codes`	The public API error codes of Appendix A (`code` , `status` , `description`)
`x-test-cards`	The test cards of chapter 12 (`number` , `outcome` , `description`)
`x-rate-limit`	`{ limit: 60, windowSeconds: 60, scope: "merchant × environment" }`

The language of the descriptions can be selected with `?lang=ja` / `?lang=en` ; otherwise `x-lang` / `Accept-Language` (default `ja`). The response carries `Cache-Control: public, max-age=300` . No authentication is needed.

5.10 Not implemented / planned (chapter 5)

- No API endpoint captures, cancels or expires a transaction.
- No API reads, lists or cancels refund requests; no API reads, lists or disables payment links; no customer, stored-card or subscription API.
- `GET /transactions` does not validate `status` server-side; an unknown value returns zero rows.
- No API retrieves or rotates webhook signing secrets (only the portal's test-key rotation).

---

6. Webhook specification

When a transaction, refund, chargeback or subscription changes state, Wire Pay sends an HTTP request (webhook) to the merchant's notification URL (**`notifyUrl`**). Delivery is implemented in `lib/webhooks/deliver.ts`, payload construction in `lib/merchants/notification.ts`. Deliveries are visible in the portal's "Webhook log" (4.7) and on the PSP dashboard.

6.1 Events

`event`	When	**`status`** value	Transaction (**`payment_id`**)
`payment.completed`	The transaction completed (immediate card / QR completion, the PSP's "Complete", an exact / excess deposit, hosted-page completion, matching an unmatched deposit, a late deposit applied through `late_deposit`, API re-notify)	`captured`	The transaction
`payment.failed`	The transaction failed (PSP "Fail", provider webhook)	`failed`	The transaction
`payment.expired`	Expired past the transfer / payment deadline (including the hourly cron)	`expired`	The transaction
`payment.cancelled`	The transaction was cancelled	`cancelled`	The transaction
`refund.completed`	A refund request was approved and completed (including a direct refund via a provider webhook)	`partially_refunded` / `refunded`	The original transaction
`chargeback.opened`	A chargeback was registered	`chargeback:<reason code>` (e.g. `chargeback:fraud`) — a fixed string with the reason, not the transaction status. `amount` is the disputed amount	The original transaction

event	When	**status** value	Transaction (**payment_id**)
limit.exceeded	A transaction was recorded as failed because of the merchant's limit	failed:limit_exceeded:<per_transaction|daily|monthly>	The failed transaction
subscription.charged	A recurring charge succeeded	captured	The charge's transaction
subscription.payment_failed	A recurring charge failed and awaits a retry	past_due	The failed transaction
subscription.suspended	The subscription was suspended after the third failure	suspended	The failed transaction

- No delivery for transactions without a merchant (provider-direct transactions) or to merchants with an empty notifyUrl (no delivery row is created either).
- Creation-time rejections send no **payment.failed**: a limit hit sends only limit.exceeded; a blacklist hit sends nothing.
- A successful recurring charge sends only subscription.charged, not payment.completed (createTransaction suppresses it for source = subscription).

6.2 Notification settings (three choices per merchant)

The content is determined by the merchant's notification settings (PSP dashboard, merchant detail "Notifications" tab).

(1) Format (**notifyFormat**) — transport

Value	Label	Method	Content-Type	Where the parameters go	Signed payload
json_post (default)	JSON (POST)	POST	application/json	JSON body (amount numeric, others strings)	The request body as sent
form_post	form-urlencoded (POST)	POST	application/x-www-form-urlencoded	Body	The request body as sent
query_get	Query parameters (GET)	GET	none	URL query string	The query string (without the leading ?)

(2) Parameter set (**notifyParamSet**) — cumulative

Value	Label	Parameters
minimal	Minimal	Identifiers + event , merchant_id , amount , currency , paid_at , status
standard (default)	Standard	Minimal + payment_method , user_id , product_id
full	Full	Standard + domain , transaction_id (always, regardless of the identifier setting)

(3) Identifier format (**`notifyIdFormat`**)

Value	Label	Identifiers included
`payment_id` (default)	Unique payment ID only	`payment_id`
`transaction_id`	Transaction ID only	`transaction_id`
`both`	Both	`payment_id` , `transaction_id`

6.3 Parameters

Parameter order is fixed: identifiers → minimal → standard → full. Because the order matters for form / query signature verification, receivers must sign the body / query string exactly as received (never re-serialise).

Name	Type	Description
`payment_id`	string	Wire Pay's transaction ID (`id` in the API)
`transaction_id`	string	The provider-side transaction ID (`Payment.externalId`); empty when unknown. In the demo a value shaped `<first 6 chars of providerKey>_<time base36>_<random>`
`event`	string	Event name of 6.1. Branch on this first
`merchant_id`	string	Merchant ID (`MER-00001`)
`amount`	integer / string	Amount in yen; numeric in JSON, string in form / query. The transaction amount even in refund / subscription events (`chargeback.opened` alone carries the disputed amount)
`currency`	string	Always `JPY`
`paid_at`	string	Completion time (`capturedAt`); for incomplete transactions the last update. ISO 8601 / UTC (e.g. `2026-09-01T03:34:56.000Z`)
`status`	string	Table in 6.1
`payment_method`	string	`credit` / `qr` / `convenience` / `bank` / `other` (may be empty)
`user_id`	string	The merchant's customer id (`customerId`); for payment links the e-mail or `guest_<time base36>`
`product_id`	string	Product name (`productName`)
`domain`	string	Host name of the notification URL

6.4 Headers and signature

Header	Content
`X-Timestamp`	Send time (UNIX seconds, string)
`X-Signature`	`v1=<signature (64 hex chars)>`
`Wirepay-Signature`	Compatibility: `t=<X-Timestamp>,v1=<same signature>`
`Content-Type`	POST only (6.2)
`User-Agent`	`WirePay-Notify/1.0`

Signature:

```
signed_payload = "<X-Timestamp>" + "." + <payload>      # payload: body, or the GET query string without "?"
v1 = hex( HMAC_SHA256( key = signing key, message = signed_payload ) )
```

The signing key is the merchant's signing secret (`wp_<test\|live>_ss_…`) of the same environment (test / live): the most recently issued one (`issuedAt` descending) that is `apiKeyUsable` (neither revoked nor expired).

- No webhook is sent to an environment without a usable signing secret (fail closed, 1.1.0; earlier versions fell back to the merchant password). The delivery is logged as `failed` with `error: signing_secret_missing`, and the PSP notification centre gets a warning (`webhook.signing_secret_missing`, once per 24 hours per merchant × environment). After a signing secret is issued a manual resend signs and sends it.
- The test signing secret is issued automatically at merchant registration (3.1); the live one is issued by the PSP at go-live.

Because the key differs per environment, keep both the test and the live signing secret and verify according to the environment of the transaction (※ the payload carries no environment field; notifications for transactions created with a test key are signed with the test secret).

Verification steps

1. Take `X-Timestamp` and the `v1` value of `X-Signature`.
2. Recompute `v1` over the body (or the GET query string) exactly as received and compare in constant time.
3. Reject if `X-Timestamp` differs from the current time by 5 minutes or more (replay protection; resends carry a fresh timestamp and signature).
4. On success branch on `event` and `payment_id` and answer 2xx.

Sample code (Node.js)

```
import { createHmac, timingSafeEqual } from "node:crypto";

export function verify(headers, rawPayload, secret) {
  const ts = headers["x-timestamp"];
  const v1 = String(headers["x-signature"] ?? "").replace(/^v1=/, "");
  if (!ts || !v1 || Math.abs(Date.now() / 1000 - Number(ts)) > 300) return false;
  const expected = createHmac("sha256", secret).update(`${ts}.${rawPayload}`).digest("hex");
  return expected.length === v1.length && timingSafeEqual(Buffer.from(expected), Buffer.from(v1));
}
```

Express (JSON POST; read the body as a raw string):

```
app.post("/wirepay/notify", express.raw({ type: "*/*" }), (req, res) => {
  const raw = req.body.toString("utf8");
  if (!verify(req.headers, raw, process.env.WIREPAY_SIGNING_SECRET)) return res.status(401).end();
  const n = JSON.parse(raw);
  // when n.event === "payment.completed", mark order n.payment_id as paid (idempotently)
  res.status(200).end();
});
```

PHP:

```
$ts  = $_SERVER['HTTP_X_TIMESTAMP'] ?? '';
$v1  = preg_replace('/^v1=/', '', $_SERVER['HTTP_X_SIGNATURE'] ?? '');
$raw = $_SERVER['REQUEST_METHOD'] === 'GET' ? $_SERVER['QUERY_STRING'] : file_get_contents('php://input');
$ok  = $ts !== '' && abs(time() - (int)$ts) <= 300
    && hash_equals(hash_hmac('sha256', "$ts.$raw", $secret), $v1);
```

6.5 Sample (JSON / standard / unique payment ID only)

```
POST https://shop.example.jp/wirepay/notify
Content-Type: application/json
X-Timestamp: 1756697696
X-Signature: v1=3f0c9c5d...
Wirepay-Signature: t=1756697696,v1=3f0c9c5d...
User-Agent: WirePay-Notify/1.0

{
  "payment_id": "clx1a2b3c4d5e6f7g8h9i0j1k",
  "event": "payment.completed",
  "merchant_id": "MER-00001",
  "amount": 4980,
  "currency": "JPY",
  "paid_at": "2026-09-01T03:34:56.000Z",
  "status": "captured",
  "payment_method": "credit",
  "user_id": "u_10021",
  "product_id": "Spring jacket"
}
```

With form-urlencoded (POST) the body is `payment_id=…&event=payment.completed&merchant_id=MER-00001&amount=4980&…` ; with query GET the same string follows `?` in the URL. The PSP dashboard's notification tab renders a live sample (with a curl command) for the chosen settings.

6.6 Delivery, timeout and retries

Item	Specification
Trigger	After the state change is committed (a failing webhook never fails the operation)
Timeout	10 seconds (`WEBHOOK_TIMEOUT_MS`). Redirects are not followed (`redirect: "manual"` ; a 3xx counts as failure)
Success	HTTP 2xx
Automatic retries	After a failure (non-2xx, timeout, network error): 1 min → 5 min → 30 min → 2 h → 24 h (`RETRY_DELAYS_MS`). Due retries are sent by the cron `/api/cron/webhook-retry` (up to 100 per run; the demo's `vercel.json` runs it once a day at 01:30 UTC so retries are effectively daily, hourly is recommended in production — Specification B 8.4)
No signing key	Deliveries to an environment without a usable signing secret are not sent and are logged as `failed` (`error: signing_secret_missing`) (6.4, 1.1.0)

Item	Specification
Abandonment	After 5 failures (`MAX_ATTEMPTS`) the delivery becomes `abandoned` and the PSP notification centre gets a critical notice. A manual resend revives it
Delivery statuses	`pending` / `success` / `retrying` / `failed` / `abandoned`
Records	One `WebhookAttempt` per try (attempt number, response code, first 1 KB of the response body, duration, trigger = `system` / `cron` / `dashboard`)
Manual resend	From the portal (4.7) or the PSP dashboard. Timestamp and signature are recomputed and the request is sent immediately regardless of schedule

6.7 Receiver requirements

Requirement	Content
Endpoint	`https` only (validated on the PSP dashboard); normally reachable from the internet
Response	Return 2xx promptly (within 10 seconds) after verification; do heavy work asynchronously
Idempotency	The same `payment_id` + `event` may arrive more than once (automatic retry, manual resend, API re-notify). De-duplicate on `payment_id` + `event`
Ordering	Several events may arrive for one transaction (e.g. `payment.completed` → `refund.completed`) and order is not guaranteed. Decide from your own state and `event`, not from `paid_at` / `status` recency
Verification	When in doubt, `GET /api/v1/transactions/{id}` returns the current state
Key management	Keep the per-environment signing secrets safe; during a rotation accept both old and new keys. Revoking a signing secret without issuing a new one stops deliveries for that environment (6.4)
IP allow list	Wire Pay's source IP is not fixed (serverless). Verify by signature, not IP (※要確認: the production egress range depends on the hosting environment)

6.8 Requesting a resend

1. The Resend button in the portal's "Webhook log" (own deliveries, immediate).
2. For a completed bank transfer whose notification was missed, ask PSP support for an API re-notify (PSP dashboard, transaction detail "API re-notify" → resends `payment.completed`; completed / partially refunded / refunded bank transfers only).
3. Abandoned deliveries (5 failures) can be resent by either method.

6.9 Not implemented / planned (chapter 6)

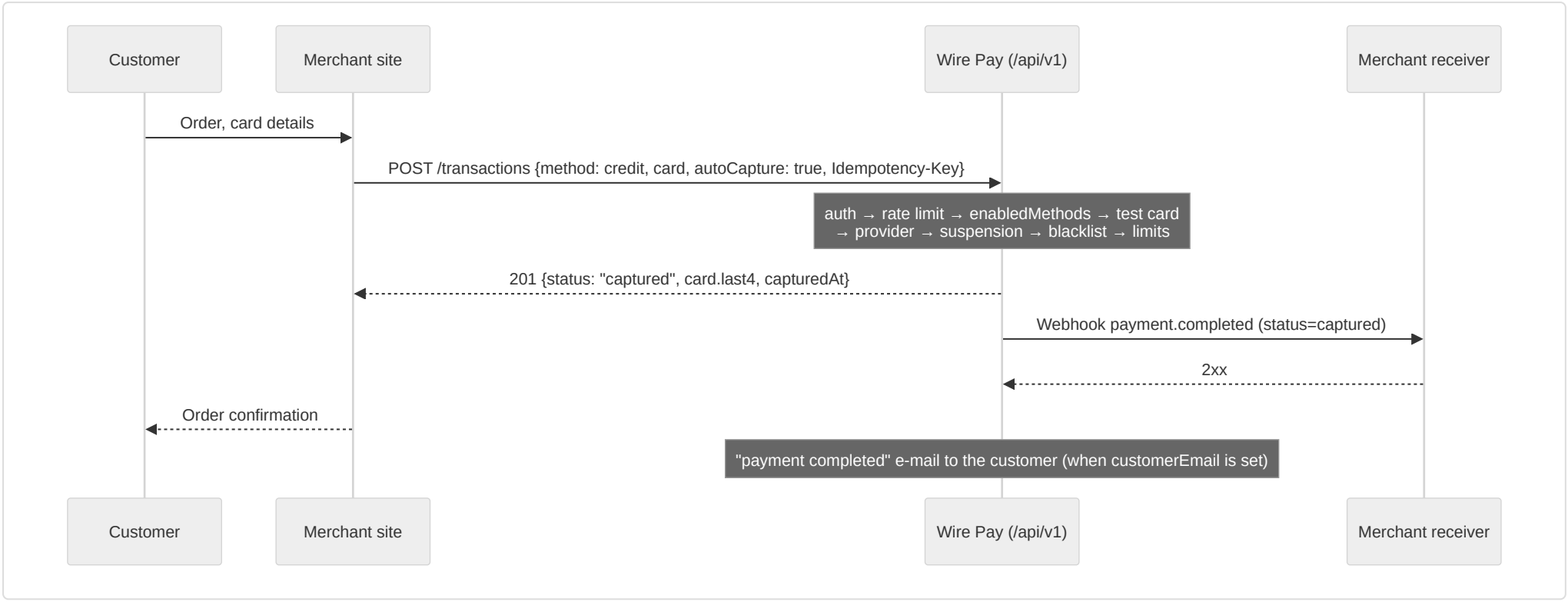
- Merchants cannot change their notification URL or format themselves (ask the PSP).
- No per-event subscription settings; every event goes to the same URL.
- The payload has no environment (test / live) field.

---

7. Payment flows in detail

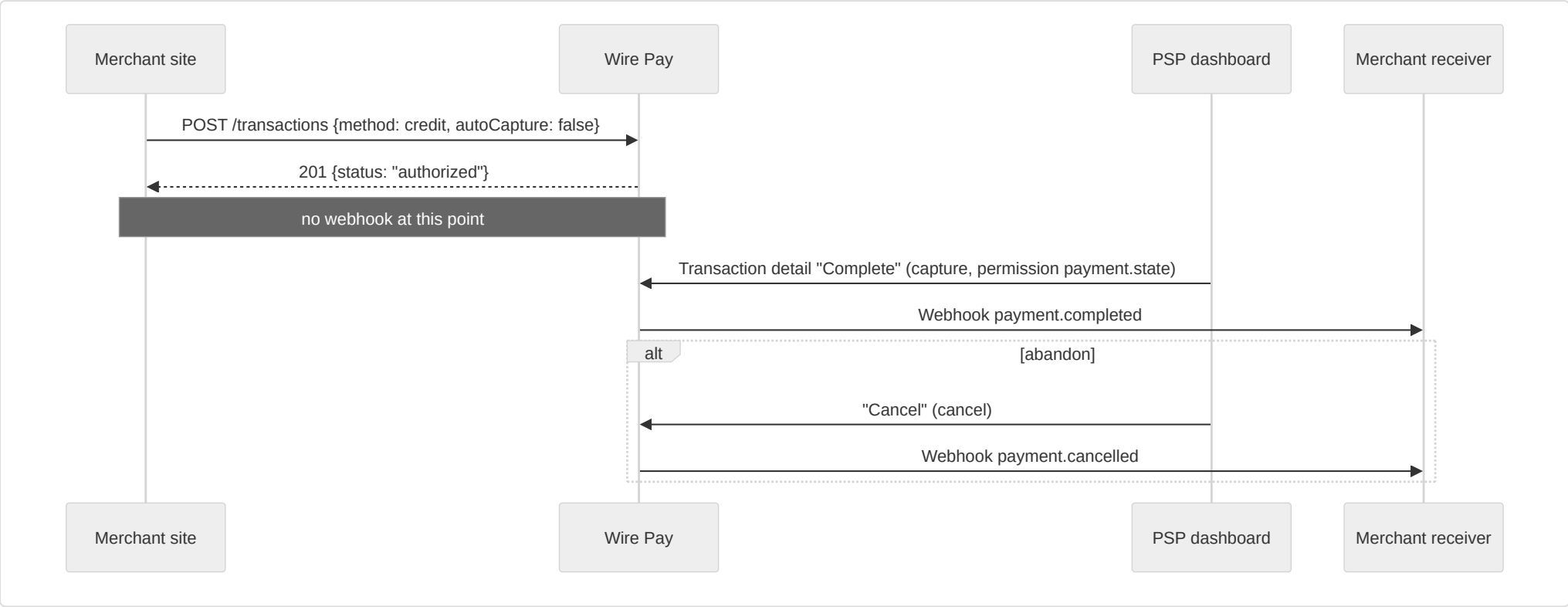
This chapter shows, per payment method, the flow from creation to completion, webhook and display as sequence diagrams. API fields are in chapter 5, webhook fields in chapter 6, statuses in chapter 2.

7.1 Credit / QR (immediate completion)



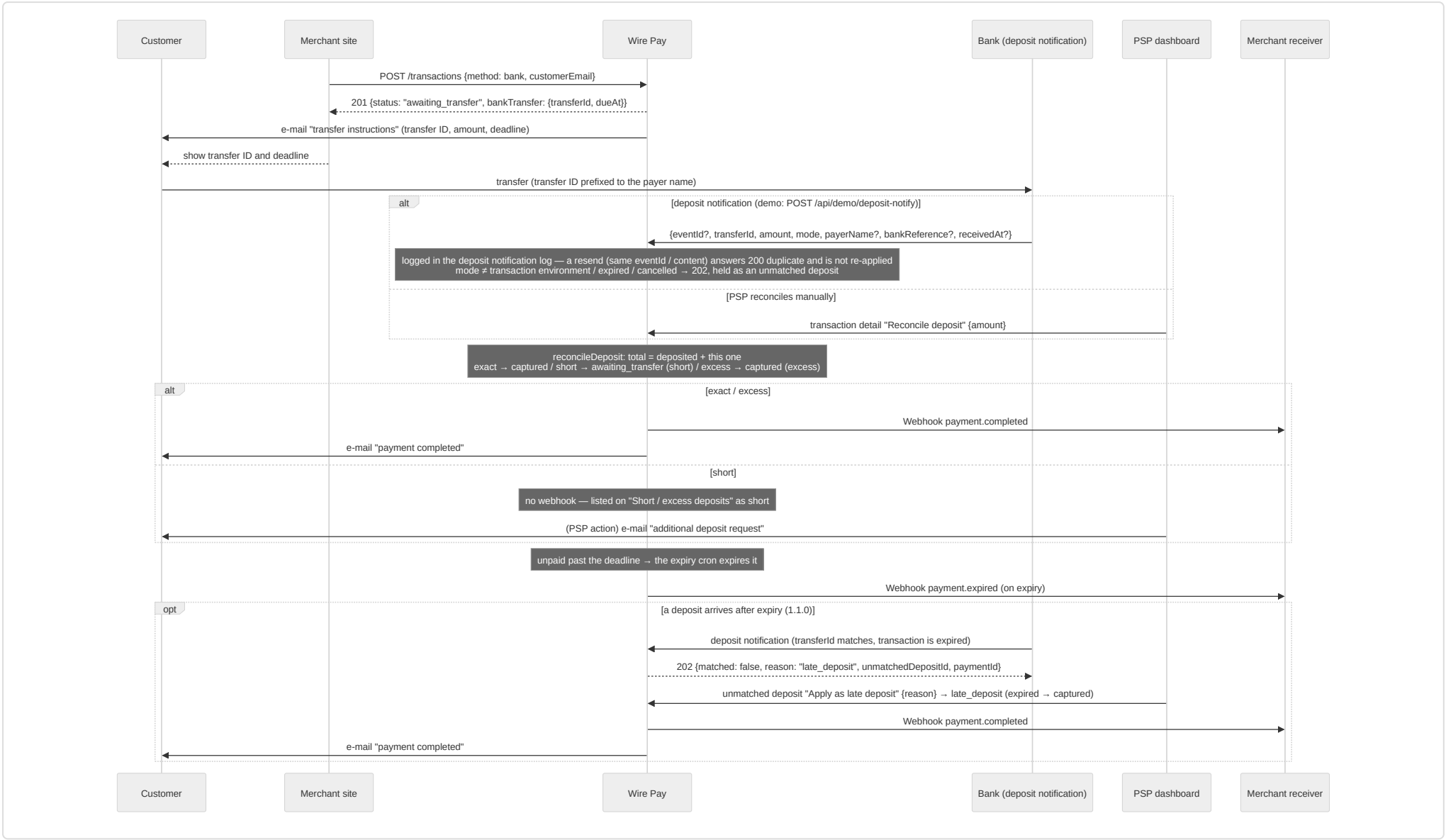
- In the demo, authorization and capture happen together and the transaction is created `captured` (the history records `created` (→ authorized) and `capture` (authorized → captured)).
- `payment.completed` is delivered almost simultaneously with the creation response. The response `status` already confirms completion; waiting for the webhook is unnecessary.

7.2 Credit / QR (authorization only → PSP completes)



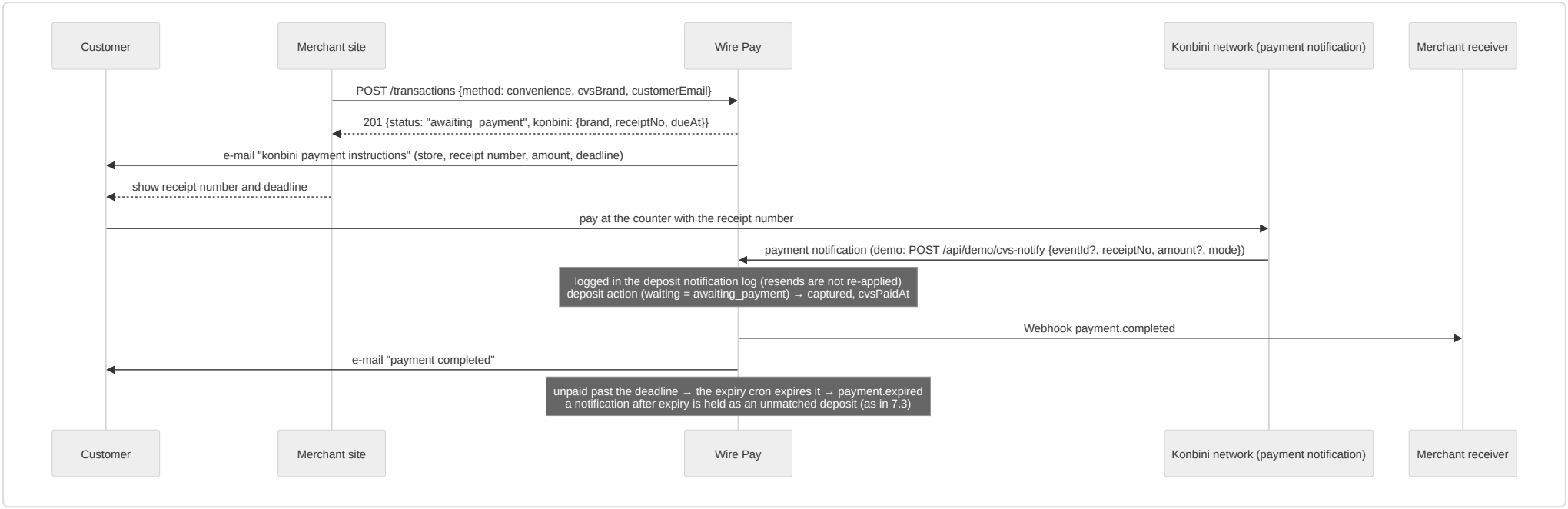
- There is no merchant API to capture or cancel an `authorized` transaction. Test card `4000000000003220` reproduces this state.

7.3 Bank transfer



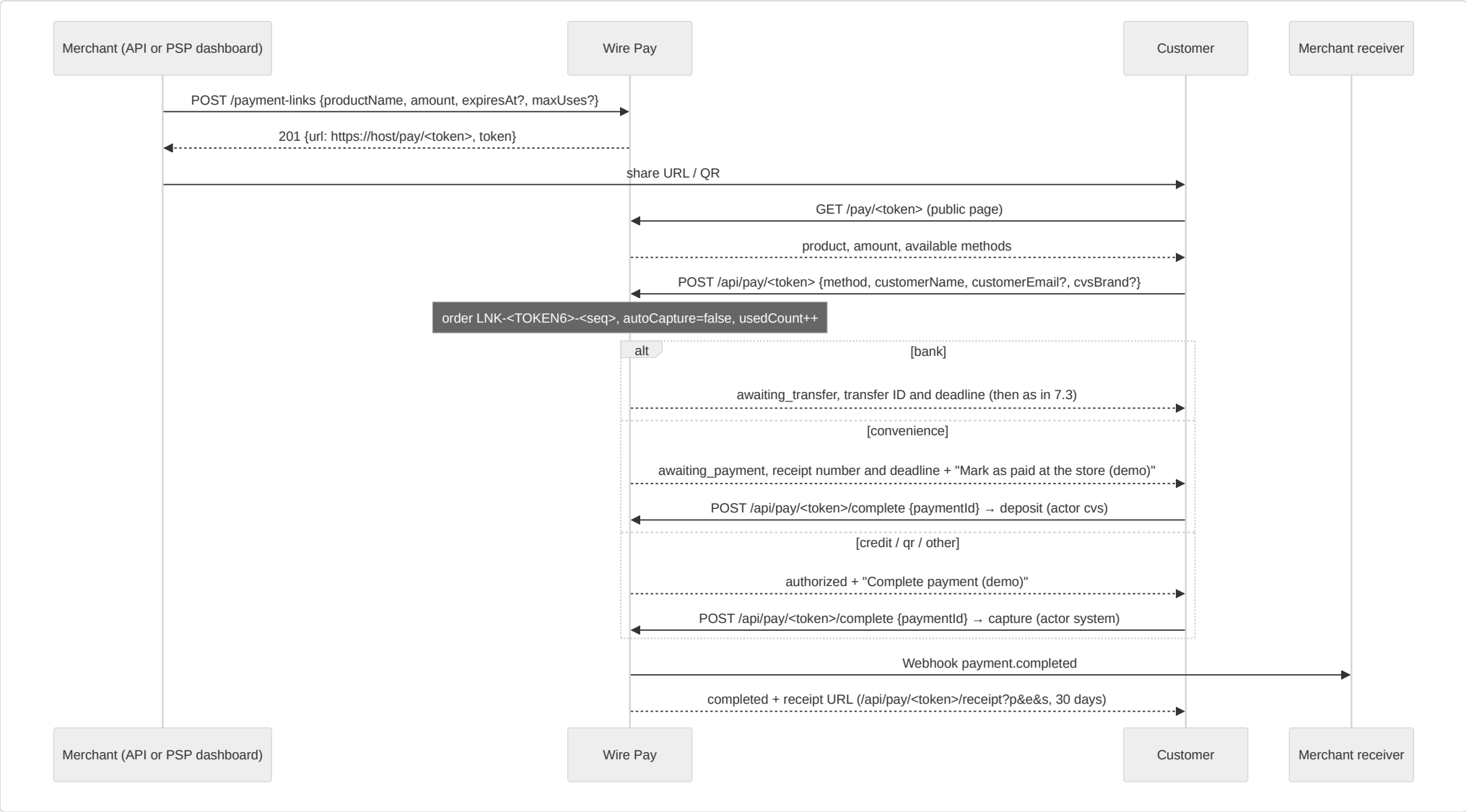
- Deposits without or with a wrong transfer ID are held as "unmatched deposits"; PSP support attaches the correct transfer ID, the same reconciliation runs, and `payment.completed` is delivered on completion. No merchant action is needed.
- For an excess deposit the PSP files a refund request (reason `duplicate`); `refund.completed` follows approval.
- Late deposits (arriving after the transaction is `expired`) and environment mismatches (e.g. a test notification whose transfer ID matches a live transaction) are held as unmatched deposits too (1.1.0). A late deposit applied by the PSP with a reason runs the `late_deposit` transition to `captured`; it completes even when the amount differs and the difference is recorded in `depositDiff`. A deposit on a cancelled transaction cannot be applied; it is dismissed as a return (refund arranged outside the system).
- A notification the bank re-sends (same `eventId`, or same source / transfer ID / amount / minute of receipt / bank reference) is never counted twice (1.1.0). Every notification is kept in the PSP dashboard's "Deposit notification log".

7.4 Konbini


Customer
Merchant site
Wire Pay
Konbini network (payment notification)
Merchant receiver
POST /transactions {method: convenience, cvsBrand, customerEmail}
201 {status: "awaiting_payment", konbini: {brand, receiptNo, dueAt}}
e-mail "konbini payment instructions" (store, receipt number, amount, deadline)
show receipt number and deadline
pay at the counter with the receipt number
payment notification (demo: POST /api/demo/cvs-notify {eventId?, receiptNo, amount?, mode})
logged in the deposit notification log (resends are not re-applied)
deposit action (waiting = awaiting_payment) → captured, cvsPaidAt
Webhook payment.completed
e-mail "payment completed"
unpaid past the deadline → the expiry cron expires it → payment.expired
a notification after expiry is held as an unmatched deposit (as in 7.3)
Customer
Merchant site
Wire Pay
Konbini network (payment notification)
Merchant receiver

- The receipt number has 13 digits (displayed 4-4-5, plain in the API). The notification accepts hyphens.
- A short notification leaves the transaction `awaiting_payment` with `depositDiff = short` (same handling as bank transfers).

7.5 Payment links (hosted payment page)

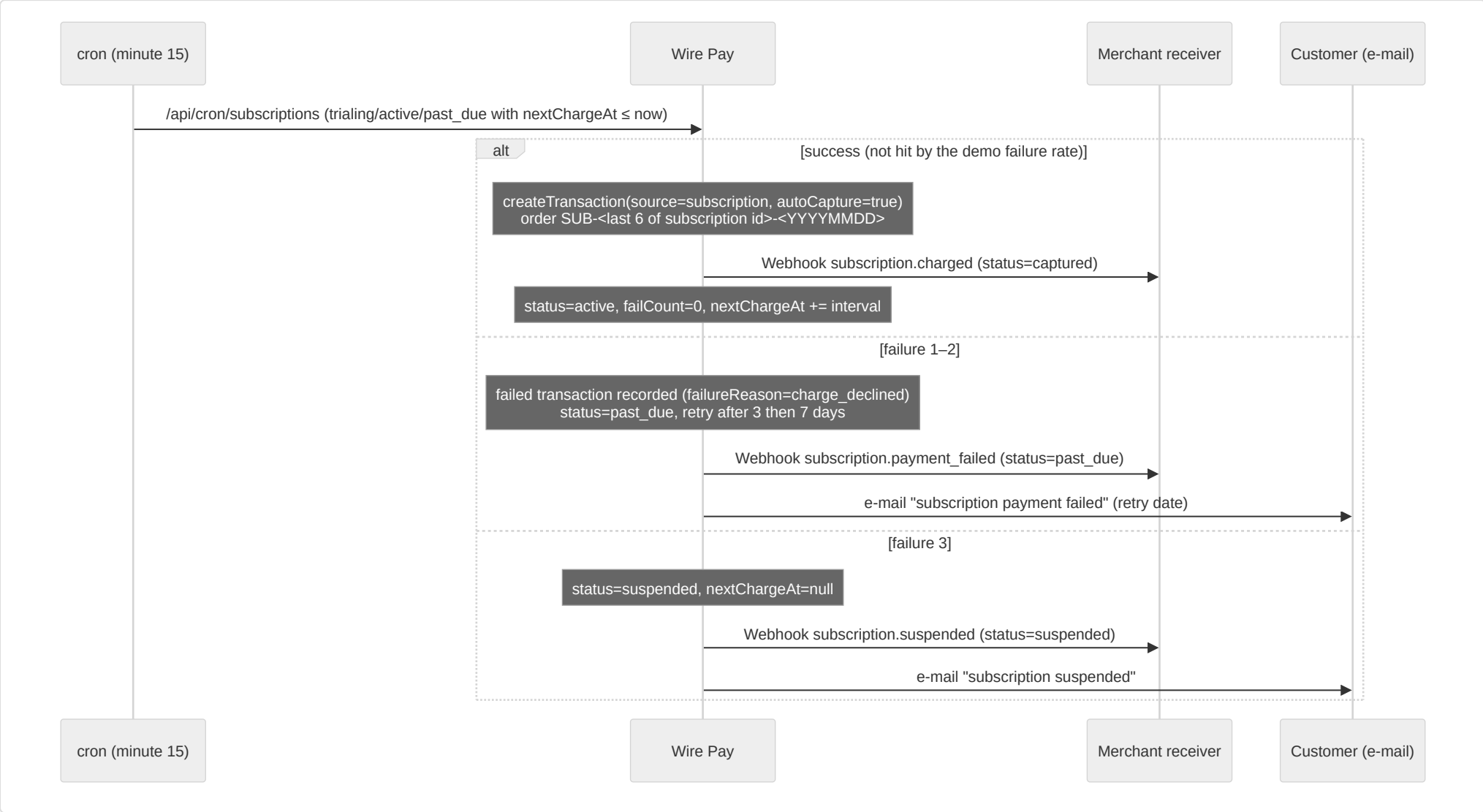


Item	Specification
Available methods	The merchant's `enabledMethods` ∩ not suspended ∩ an active provider exists in the same environment
Order number	`LNK-<first 6 token chars (upper case)>-<uses + 1, 4 digits zero-padded>`
Customer id (`user_id`)	The customer's e-mail; `guest_<time base36>` when none
Unavailability order	`disabled` (disabled) → `expired` (`expiresAt` ≤ now; midnight JST after the given day) → `exhausted` (`usedCount` ≥ `maxUses`). Checked on page load and on POST (POST answers 409 `link_disabled` / `link_expired` / `link_exhausted`)
Uses	+1 per transaction creation (consumed at creation, not completion)
Receipt	A signed URL is minted on completion (`HMAC-SHA256(token.paymentId.exp)`, 30 days). The first download creates the receipt row (`source = customer`, number `RC-YYYYMM-00001`); later downloads are not counted as re-issues
Audit	Transactions created from the page use actor `system`; completion uses `system` (`cvs` for konbini)

Disabling a payment link is PSP-dashboard only (`/payment-links/[id]`).

7.6 Subscriptions

Plans and subscriptions are created only on the PSP dashboard (`/subscriptions`). Merchants receive the results by webhook.



Item	Specification (**`lib/subscriptions/service.ts`**)
Interval	`monthly` (calendar month + 1) / `yearly` (+ 1 year) / `weekly` (+ 7 days)
Trial	`trialDays` > 0 starts `trialing` ; the trial end is the first charge date
Retries	`RETRY_DAYS = [3, 7]` , `MAX_CHARGE_FAILURES = 3`
Provider	The active credit provider of the same environment (else any active provider)
Demo failure rate	`RiskSetting.demoChargeFailPercent` (default 20 %). The PSP's "Charge now (success / failure)" forces the outcome
Charge transactions	Ordinary transactions in lists, refunds, settlement and reports; linked by `subscriptionId`

7.7 Not implemented / planned (chapter 7)

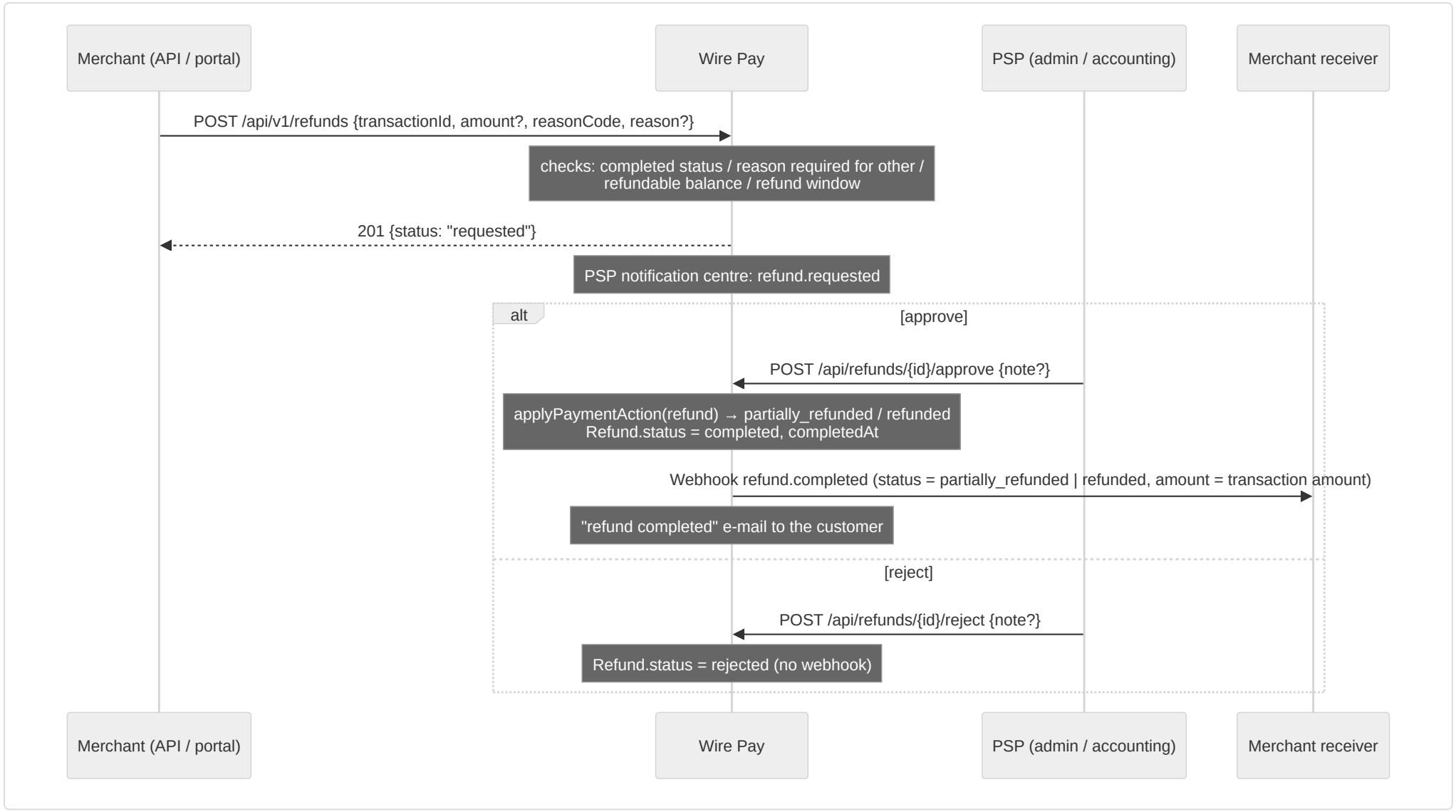
- No merchant API for plans / subscriptions.
- The payment page has no real card entry, 3-D Secure or QR display ("Complete payment (demo)" completes it).
- The payment page has no button simulating a bank deposit (PSP reconciliation or the demo deposit API).

---

8. Refunds, voids and chargebacks

8.1 Refunds

A refund is a request row in the **`Refund`** table with a two-step flow "request → approve / reject". Requests come from merchants (API / portal) and PSP operators; approval / rejection is PSP-only (admin / accounting).


Merchant (API / portal)
Wire Pay
PSP (admin / accounting)
Merchant receiver
POST /api/v1/refunds {transactionId, amount?, reasonCode, reason?}
checks: completed status / reason required for other / refundable balance / refund window
201 {status: "requested"}
PSP notification centre: refund.requested
alt
[approve]
POST /api/refunds/{id}/approve {note?}
applyPaymentAction(refund) → partially_refunded / refunded
Refund.status = completed, completedAt
Webhook refund.completed (status = partially_refunded | refunded, amount = transaction amount)
"refund completed" e-mail to the customer
[reject]
POST /api/refunds/{id}/reject {note?}
Refund.status = rejected (no webhook)
Merchant (API / portal)
Wire Pay
PSP (admin / accounting)
Merchant receiver

Request validation (`createRefundRequest`)

Condition	Result
Transaction not `captured` / `partially_refunded`	409 `invalid_transition`
`reasonCode = other` without `reason`	400 `invalid_input` (`fieldErrors.reason = required`)
`amount` > refundable balance = amount − refunded − pending requests (`requested` / `approved` / `processing`)	400 `refund_exceeds`
Past the refund window (`capturedAt` + the merchant's `refundWindowDays`, default 180 days, 0–365)	409 `refund_window_expired`. Only a PSP admin may request with "forced refund" (`force: true` + reason required); never through merchant paths
Kind	`amount` ≥ transaction amount → `full`, else `partial`

Refund request statuses

Status	Meaning
`requested`	Just filed; awaiting review
`approved`	Approved (skipped in the demo — goes straight to `completed`)
`processing`	In progress (not used in the demo)
`completed`	Refund completed; the transaction is `partially_refunded` / `refunded`
`rejected`	Rejected; `decisionNote` holds the reviewer's note (shown in red in the portal)
`failed`	Failed (does not occur in the demo)

What the merchant sees

Where	Content
Transaction detail (portal)	Refund requests (reason, time, amount, status), refunded total, refundable balance
Refund requests (portal)	All requests and outcomes, rejection reasons
Webhook	`refund.completed` (approval only; rejections are not notified)
API	Only the `POST /api/v1/refunds` response; there is no refund read API (chapter 14)
Settlement	Completed refunds are deducted in the batch of the period containing `completedAt` (chapter 9)

Multiple refunds

Partial refunds may be requested repeatedly within the balance. Once the refunded total reaches the amount the transaction is `refunded` and further requests answer 409 `invalid_transition`.

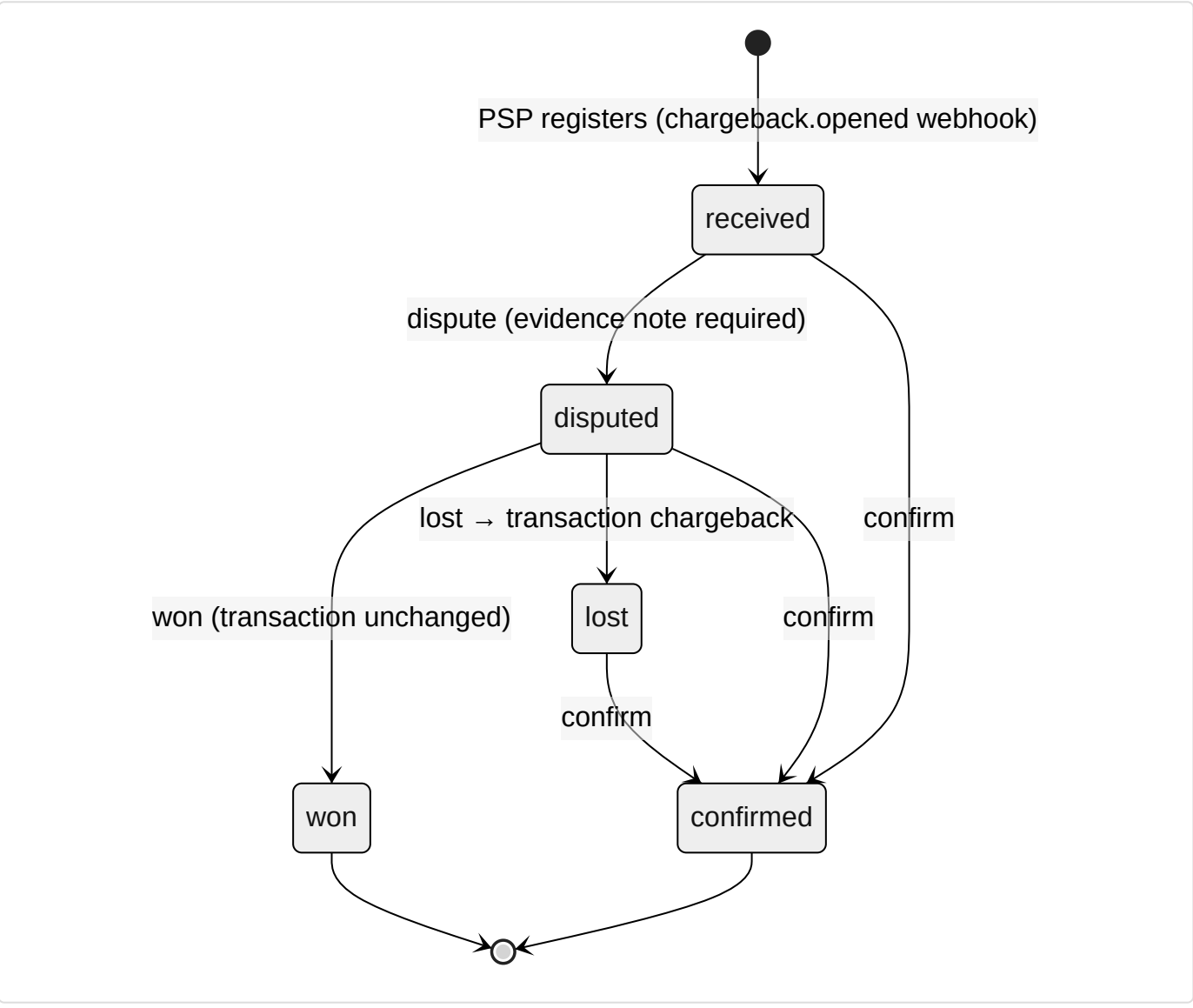
8.2 Voids (cancel)

Abandoning an `authorized`, `pending` or `awaiting_transfer` transaction before completion. Distinct from refunds; not available after completion.

Item	Specification
Actor	PSP only (permission `payment.state` = admin). Not available through the merchant API or portal
Result	`cancelled`, `cancelledAt`, webhook `payment.cancelled` (`status = cancelled`)
Settlement	Not settled (never completed)
How to request	Tell PSP support the transaction ID (or order number) (※要確認: the request channel is outside the system)

8.3 Chargebacks

Payment disputes from card issuers are registered and managed by the PSP. Merchants learn about them from the registration webhook and, when lost / confirmed, from the transaction status and the settlement deduction.



Item	Specification (`lib/chargebacks/*`)
Eligible transactions	`captured` / `partially_refunded` / `refunded`. A second open case (`received` / `disputed`) answers 409 `chargeback_exists`
Registration	Disputed amount (default = transaction amount, ≤ amount), reason code (`fraud` / `not_received` / `not_as_described` / `duplicate` / `other`), dispute deadline (default 14 days), evidence note

Item	Specification (**`lib/chargebacks/*`**)
Webhook	`chargeback.opened` at registration (`status = chargeback:<reason>` , `amount` = disputed amount). Won / lost / confirmed are not notified
Transaction	Lost (`lost`) or confirmed (`confirmed`) moves the transaction to `chargeback` (transitions in 2.3)
Settlement	Lost / confirmed amounts are deducted in the batch of the period containing **`resolvedAt`**
Deadline	The PSP notification centre warns daily from 3 days before the deadline
Evidence	Outside the system (contact the PSP). The portal has no chargeback screen

8.4 Not implemented / planned (chapter 8)

- No merchant API / portal void, refund cancellation or read, chargeback read or evidence upload.
- No webhooks for refund rejection or chargeback won / lost / confirmed.
- Refund statuses `approved` / `processing` / `failed` are unused in the demo (approval completes immediately).

---

9. Settlement

Settlement is managed as settlement batches (**`SettlementBatch`**) per merchant and period with their lines (**`SettlementLine`**). The PSP (admin / accounting) generates, confirms and pays batches; merchants read them in the portal (4.8 / 4.9) and through the merchant API (5.8). Implementation: `lib/settlements/*` ; period and payout dates use midnight JST as boundaries.

9.1 Settlement settings (per merchant)

Set on the PSP dashboard, merchant detail "Settlement" tab.

Item (column)	Options / constraints	Meaning
Settlement cycle (`settlementCycle`)	`monthly` month-end (default) / `semimonthly` 15th and month-end / `weekly`	Default period of a batch
Payout term (`payoutTerm`)	`next_month_end` (default) / `next_month_15` / `month_after_next_15` / `next_week`	Payout date computed from the period end
Bank account	Bank (≤80), branch (≤80), type `ordinary` / `current` , account number (7–8 digits), holder (≤80)	Payout destination
Settlement fee (`settlementFee`)	0–10,000 yen (default 0)	Transfer fee deducted once per batch
Registration number (`invoiceNumber`)	`T` + 13 digits	Printed on the qualified invoice as the recipient
Address (`address`)	≤200	Same

9.2 Cycles and periods

Cycle	Default period (relative to the generation date)	Example (generated 3 Sep)
Month-end	1st to last day of the previous month	1–31 Aug
15th and month-end	Generated on/after the 16th: 1st–15th of this month; on/before the 15th: 16th–end of the previous month	16–31 Aug
Weekly	Previous Monday–Sunday	24–30 Aug

- The PSP may choose any period (start ≤ end). Internally the end is stored as midnight JST of the next day, exclusive (`periodTo` in the API is exclusive).
- One batch per merchant, environment and identical start/end (duplicates answer 409 `settlement_exists`).

Payout term	Scheduled payout (**`scheduledAt`**, from the period end)
Next month end	Last day of the following month
Next month 15th	15th of the following month
Month after next 15th	15th of the month after next
Next week	7 days after the period end

9.3 Line aggregation (**`collectLines`**)

Kind	Source	Date used	Amount	Fee	Tax category
`payment` sale	Transactions with status `captured` / `partially_refunded` / `refunded` / `chargeback`	`capturedAt` within the period	Transaction amount (+)	Formula below	`standard10`
`refund`	Refund requests with status `completed`	`completedAt` within the period	Refund amount (−)	0	`exempt`
`chargeback`	Chargebacks with status `lost` / `confirmed`	`resolvedAt` within the period	Disputed amount (−)	0	`exempt`

- Refunds and chargebacks are deducted in the period in which they occurred, whatever period the original sale belonged to. The net payout may be negative.
- Batches are per environment (test / live); in production test batches are never paid (demo only).

9.4 Fee formula

```
fee (per transaction) = floor( amount × rate% ÷ 100 ) + fixed fee (yen)      # rounded down to the yen
```

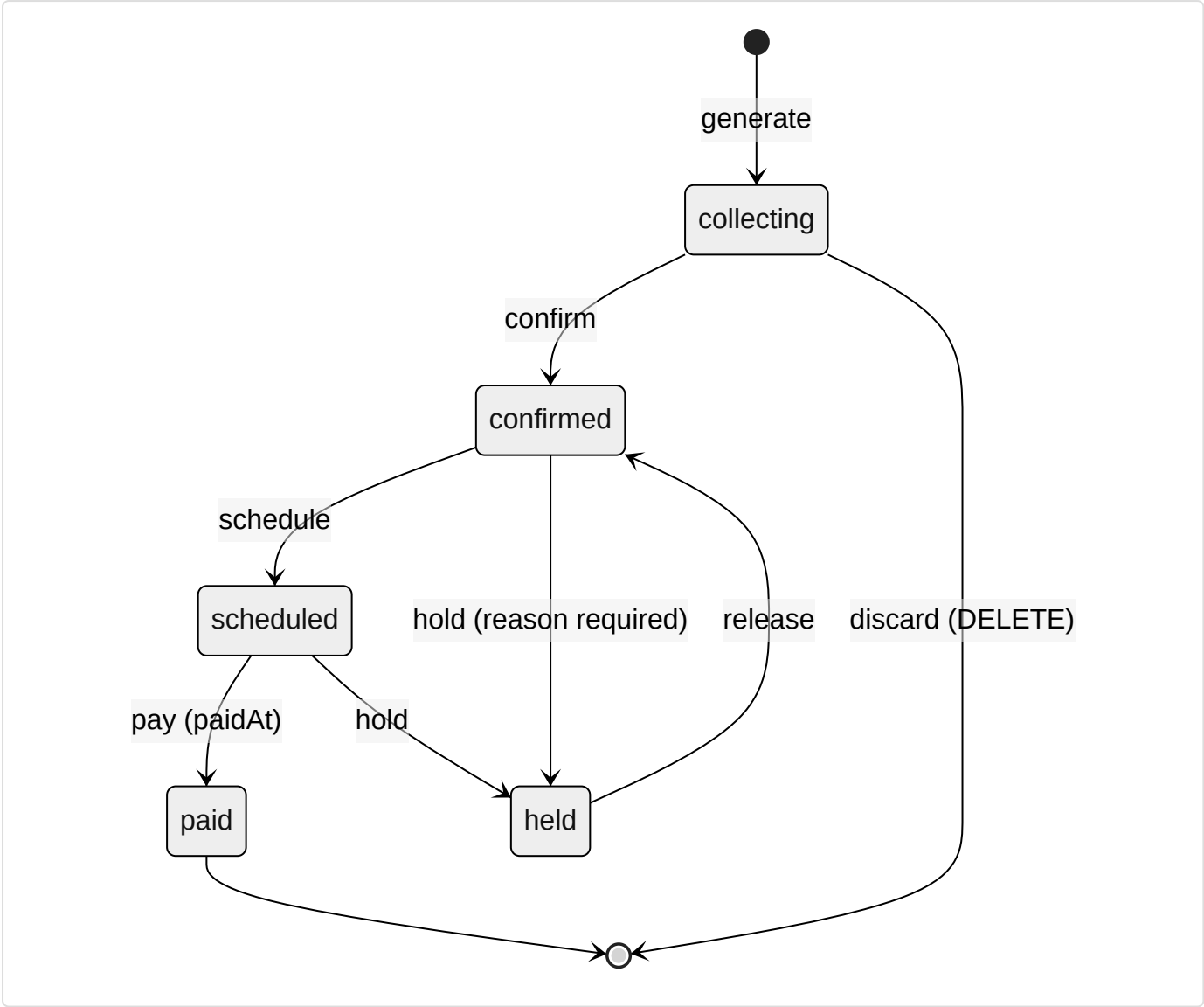
- Rate and fixed fee come from the contract in force at the transaction's completion time (the latest schedule in the contract history with `effective date ≤ capturedAt`), per payment method.
- No contract, or method `other` → 0 yen.
- Default plan rates (`PLAN_DEFAULT_RATES`):

Plan	Credit	QR	Konbini	Bank transfer
Standard	3.6 % + 0 yen	3.0 % + 0 yen	3.5 % + 30 yen	2.0 % + 50 yen
Premium	3.2 % + 0 yen	2.6 % + 0 yen	3.0 % + 30 yen	1.5 % + 50 yen
Custom	Rate (0–100 %, 2 decimals) and fixed fee (0–100,000 yen) per method			

9.5 Batch summary and net payout

```
gross sales        = Σ amounts of sale lines
fee total          = Σ fees of sale lines
refund total       = Σ |amounts of refund lines|
chargeback total   = Σ |amounts of chargeback lines|
settlement fee     = the merchant's settlementFee (once per batch)
net payout (netAmount) = gross sales − fee total − refund total − chargeback total − settlement fee
```

9.6 Batch statuses and actions (PSP)



Status	Label	Meaning	Visible to the merchant
`collecting`	Collecting	Just generated; the PSP reviews the lines	List / detail (no invoice)
`confirmed`	Confirmed	Amounts fixed, lines immutable. The invoice number is assigned at confirmation (1.1.0)	Invoice number and invoice PDF available; shown as "Next payout" on the home screen
`scheduled`	Scheduled	To be paid on the scheduled date	Same
`paid`	Paid	Paid (`paidAt`)	Paid date
`held`	Held	Payout stopped, e.g. during a dispute (`holdReason`)	Hold reason (`holdReason` in the API)

9.7 Reading the statement (portal / API)

Field	Description
Period	`periodFrom` to `periodTo − 1 day` (the screen shows the inclusive end; `periodTo` in the API is exclusive)
Gross sales / fee total / refund total / chargeback total / settlement fee / net payout	9.5
Scheduled payout / paid at	`scheduledAt` / `paidAt`
Invoice number	`invoiceNo` (once confirmed; null while collecting). Searchable in the PSP's global search
Lines	Occurred, kind, order (→ transaction detail), method, amount (negative for refunds / chargebacks), fee

9.8 Qualified invoice PDF

A qualified invoice PDF from the PSP to the merchant for the payment fees of a settlement batch.

Item	Content
Availability	Batches at `confirmed` / `scheduled` / `paid` / `held`. Collecting → 409 `settlement_not_confirmed`
Paths	Portal settlement detail (`GET /api/merchant-portal/settlements/{id}/invoice`), PSP dashboard (permission `invoice.export` = admin / accounting)
Invoice number	`INV-YYYYMM-<merchant number>-<2-digit sequence>` (the period's start month; the merchant number is the merchant ID without `MER-`; the sequence counts the merchant's batches confirmed in that month). Assigned at confirmation and stored in **`SettlementBatch.invoiceNo`** (unique); the PDF prints the stored value. An unnumbered batch answers 409 `invoice_no_missing` (1.1.0; the former `INV-YYYYMM-<merchant number>` collided when several batches of one month existed under semi-monthly / weekly cycles)
Issuer	The PSP's global settings (name, registration number `T` + 13 digits, address, phone)
Recipient	The merchant's company name, address and registration number (settlement tab)
Lines	Fees per payment method (count, sales base, tax category, tax-inclusive amount) and the transfer fee
Consumption tax	Fees are tax-inclusive. Per rate: `tax = round(inclusive × rate ÷ (100 + rate))`, rounded by the global `taxRounding` (`floor` default / `round` / `ceil`). Sale-line fees are 10 %; refund / chargeback lines are exempt
Audit	`settlement.invoice_export` (`portal: true` when exported by the merchant)
PDF	pdf-lib with an embedded Noto Sans JP subset; no headless browser

9.9 Agent fees (reference)

For merchants with an agent, `floor(net payout × the agent's fee rate %)` is aggregated monthly over confirmed / scheduled / paid batches (PSP dashboard "Agent fees"). It does not change the merchant's net payout.

9.10 Not implemented / planned (chapter 9)

- No automatic generation per cycle and no payout API (the PSP generates manually; payout is a status change).
- No automatic statement e-mail (template `statement` is a manual-use text).
- No line CSV export in the portal (PSP dashboard only).

---

10. Security and limits

10.1 IP restrictions

Target (**`target`**)	Content
`merchant_api` (merchant API)	Per merchant. Callers matching a block (**`deny`**) rule receive 403 `ip_blocked` on `/api/v1/*` and `/api/merchant-api/*`. Allow (`allow`) rules are enforced when the PSP switches the allow-list mode on (below); otherwise they are informational
`admin` (dashboard)	For the PSP dashboard; not relevant to merchants (has the same allow-list mode)

Allow-list mode (`GlobalSetting.ipAllowlistMerchantApi` , default off, 1.1.0):

- When on, a merchant with at least one enabled allow rule gets 403 `ip_blocked` (`details.reason = "not_in_allowlist"`) from callers that match none of its allow rules. Merchants without allow rules are unaffected. Block rules always take precedence.
- The mode cannot be switched on without at least one enabled allow rule for the target (400 `allowlist_no_rules`). For the dashboard target (`admin`) the operator's own address must also be covered (400 `ip_self_block`), and while the mode is on, changes to allow rules that would drop the operator's own address are refused.
- Merchant action: where the allow-list mode is on, have all egress IPs of your API servers registered by the PSP before connecting.
- Rules are a single IPv4 address (`203.0.113.10`) or CIDR (`198.51.100.0/24` , `/0` – `/32`). IPv6 is not evaluated (never blocked; the IPv4-mapped form `::ffff:a.b.c.d` is treated as IPv4).
- The caller IP is the first hop of `X-Forwarded-For` , else `X-Real-IP` .
- Rules are configured by the PSP (admin / developer). Tell the PSP in advance when your API servers' egress IPs change.

10.2 Transaction limits

The PSP sets per-transaction / daily / monthly caps (yen; blank = unlimited) per merchant × method (`credit` / `qr` / `convenience` / `bank`) in `MerchantLimit` .

Item	Specification (**`lib/merchants/limits.ts`**)
When checked	At creation (API, payment links, subscriptions, PSP dashboard alike), after the blacklist check
Order	Per transaction (`amount` > cap) → daily (today's total + `amount` > cap) → monthly (this month's total + `amount` > cap)
Counted usage	Transactions of the same merchant, method and environment with status `pending` / `awaiting_transfer` / `authorized` / `captured` / `partially_refunded` / `refunded` . Failed, cancelled and expired are excluded. Day / month boundaries are JST
On exceed	The transaction is recorded as **`failed`** (HTTP 201) with `failureReason = "limit_exceeded:<per_transaction|daily|monthly>"` ; webhook `limit.exceeded` (`status = failed:limit_exceeded:…`) and a PSP notification
Merchant visibility	Failure reason on the transaction detail and the webhook. Caps and usage are on the PSP dashboard's merchant "Limits" tab (not in the portal)

10.3 Payment-method suspension

The PSP can temporarily stop new transactions only for a payment method, e.g. during a provider outage or at the merchant's request. In-flight transactions (completion, refunds, deposits) are unaffected.

Item	Specification (**`lib/payments/method-controls.ts`**)
Scope	Global (all merchants, admin) / per merchant (admin / cs). Methods: credit / QR / konbini / bank transfer
Order	Global → per merchant → the merchant's enabled methods (`enabledMethods`)
API response	409 `method_suspended` , `details: { scope: "global" | "merchant", method }`
Payment page	Suspended methods disappear from the choices
Auto-resume	A resume time can be set; the cron `/api/cron/payment-method-controls` at minute 45 resumes it
Merchant visibility	Portal "Payment methods" (reason, resume time) and any published incident announcement (the PSP can draft one when suspending globally)

10.4 Blacklist (repeated-failure detection)

Item	Specification (**`lib/risk/*`**)
Keys	Card fingerprint (`sha256(brand|last4|expiry)`), source IP (IPv4), e-mail (lower-cased)
When checked	At creation (before limits). A hit on an active entry records the transaction as `failed` (HTTP 201) with `failureReason = "blocked:<card|ip|email>"` . No webhook
Automatic detection	M failures within N minutes for the same key (default 10 minutes / 3, adjustable by the PSP) create a candidate; the PSP approves it to make it active. Candidates do not block

Item	Specification (**`lib/risk/*`**)
Manual entries	Added by the PSP (admin / cs); active immediately
Merchant visibility	The failure reason (`blocked:…`) on the transaction detail. The list itself is not exposed to merchants; ask the PSP to lift false positives

10.5 Personal data and retention

Item	Specification (**`lib/pii/*`** , **`lib/customers/service.ts`**)
Stored personal data	Customer name, e-mail, source IP, card last 4 and card fingerprint on transactions; name, e-mail and phone in the customer master
Never stored	Card number (PAN) and security code. The API's `card.number` is used only for brand detection and to derive the last 4 and the fingerprint
PSP dashboard display	Names show the family name only (`山田 * *`), e-mails the first 2 characters + `***@domain` , phones the last 4 digits, IPs hide the 4th octet. Unmasking is admin / cs only and each reveal is audited (`pii.unmask`)
Merchant portal display	Own data is unmasked
Audit log	Personal-data keys (`customerName` / `payerName` / `*email*` / `phone` / `ip` -like) are masked at write time
CSV	PSP global setting `csvPii` (`masked` default / `full`); even `full` reveals values only to admin / accounting
Retention (anonymisation)	Global `piiRetentionMonths` (default 24 months, 1–120). A daily cron (`/api/cron/anonymize` , 19:00 UTC = 04:00 JST) clears personal data of customers whose last transaction is older, and of old transactions without a customer row (name, e-mail, phone, source IP and card identifiers set to null; stored card tokens deleted). Ledger data (amounts, statuses) remains
Manual anonymisation	PSP admin with a reason (customer detail); irreversible. A customer who transacts again is re-created with fresh data and a new retention window
Audit retention	Audit rows older than `auditRetentionYears` (default 7 years, 1–30) are deleted by the same cron
Effect on merchants	Anonymised transactions / customers lose name, e-mail etc. in the portal too (transaction ID, amount, status remain); `customer` in `GET /transactions/{id}` becomes null

10.6 Customer master and stored cards

Item	Specification
Customer (`Customer`)	One row per merchant × `customerId` (the API's `customerId`). Created / refreshed at transaction creation (name and e-mail overwritten by the latest transaction)
Stored card (`PaymentToken`)	Brand, last 4, expiry and fingerprint of transactions that sent `card` ; one row per customer × fingerprint, `lastUsedAt` updated
Use	Portal "Customers" (4.13 / 4.14), PSP global search and fraud detection
API	No API reads or uses customers or stored cards. There is no re-charging with a stored card

10.7 Other transport and authentication limits

Item	Specification
Notification URL	https only
API key	Bearer only; no query-string keys
Rate limits	Public API and merchant API together 60 / minute (merchant × environment). Portal mutation APIs 60 / minute per IP, login 10 / minute
CSRF	Portal mutation APIs require a same-origin `Origin` / `Referer` and `X-Requested-With: fetch` (browser use only)
Demo caps	1,000 transactions, 200 merchants, 20 API keys per merchant, etc. (`DEMO_LIMITS`); 403 `demo_limit` when reached
Cookie	Merchant session `wirepay.merchant` : `Path=/; HttpOnly; SameSite=Lax; Max-Age=43200` , plus `Secure` in production (`NODE_ENV=production`)

10.8 Not implemented / planned (chapter 10)

- The portal cannot view or change IP restrictions, limits or the blacklist.
- Column-level encryption of personal data is not implemented (plain columns + masking + retention).
- IPv6 IP restrictions are not evaluated.

---

11. Notifications

This chapter covers notifications other than webhooks: e-mails to customers and merchants and announcements / incident information. Webhooks are in chapter 6.

11.1 The three e-mail notification settings (merchant setting **`emailNotification`**)

Merchant-facing payment mail follows the merchant setting `emailNotification` and goes to the contact e-mail (**`Merchant.email`**) (implemented in 1.1.0; the statement e-mail `statementEmail` is used for settlement matters only).

Value	Label	What is sent (template)
`per_payment`	Per payment	`merchant_payment_notice` (order number, amount, method, completed at, transaction ID) every time a transaction becomes `captured` — immediate completion, PSP "Complete", deposit reconciliation, late deposits and subscription charges alike
`daily`	Daily digest	The cron `/api/cron/merchant-daily-summary` at 09:00 JST sends one `merchant_daily_summary` per merchant × environment for the previous day 0:00–24:00 (JST): completed count and amount, breakdown by method, failed count, expired count. Nothing is sent on a day with no transactions, and a re-run never duplicates a day
`none`	None	Nothing

The current setting is shown on the portal home screen (ask the PSP to change it). In the demo the mail is only logged, as in 11.2.

11.2 E-mails to customers

For transactions with a customer e-mail (`customerEmail`), mails are produced on these events. Templates are editable by the PSP (`/email-templates` , Japanese / English); the customer language is the environment variable `EMAIL_DEFAULT_LANG` (default `ja`).

Template key	When	Main content
`transfer_guide` transfer instructions	Bank-transfer transaction created	Amount, transfer ID, deadline, product
`cvs_guide` konbini payment instructions	Konbini transaction created	Store, receipt number (4-4-5), amount, deadline
`payment_completed` payment completed	The transaction completed: `capture` , an exact / excess `deposit` , or a late deposit `late_deposit` (PSP "Complete", hosted-page completion, reconciliation / deposit notification, matching an unmatched deposit). Also sent for transactions that are **`captured`** at creation (**`autoCapture: true`** credit / QR) (1.1.0; subscription charges are excluded because they use the `subscription_*` templates)	Order number, amount, transaction ID
`refund_completed` refund completed	A refund request was approved	Refund amount, order number, transaction ID

Template key	When	Main content
`deposit_short` additional deposit request	The PSP marks a short deposit as "customer notified"	Transfer ID, shortfall, amount due, deadline
`subscription_failed`	A recurring charge failed (retry pending)	Plan, amount, retry date
`subscription_suspended`	Third failure	Plan

- The demo does not send mail; it records it in the send log (PSP dashboard "E-mail log") as "demo (not sent)" (`EMAIL_PROVIDER=log` , default). Connecting a real provider is a PSP-side configuration.
- Disabled templates produce no mail.
- Merchant-facing mail: the two templates of 11.1 (`merchant_payment_notice` / `merchant_daily_summary`) are sent automatically; `statement` (settlement statement) is a manual-use text that nothing sends automatically.

11.3 Announcements, maintenance and incidents

Item	Specification (**`lib/announcements/*`**)
Kinds	`notice` / `maintenance` / `incident`
Published when	Status `published` and `publishFrom ≤ now < publishUntil` (no `publishUntil` = until ended)
Audience	All merchants (`targetAll`) or listed merchants (`targetMerchantIds`)
Incident timeline	`occurred` → `investigating` → `resolved`
Reading (screens)	Portal "Announcements" (4.12) and the home screen's announcements panel
Reading (API)	`GET /api/merchant-api/announcements` (5.8). No recommended polling interval is defined (※要確認)
Push	No webhook or e-mail delivery of announcements

11.4 The PSP notification centre (reference)

Operator notifications (abandoned webhooks, method suspension, limit hits, refund requests, subscription failures, daily overdue digests, blacklist candidates, etc. — 17 kinds) go to the PSP dashboard's notification centre and to e-mail / Slack. Merchants do not receive them, but knowing which ones are triggered by merchant activity helps when contacting the PSP.

PSP notification	Trigger
`webhook.abandoned`	5 failed deliveries to the merchant's URL (likely a merchant endpoint problem)
`limit.exceeded`	The merchant's limit was exceeded
`refund.requested`	A refund request from the merchant

PSP notification	Trigger
`transfer.overdue` / `cvs.overdue`	Overdue transfers / konbini payments (daily)
`deposit.late`	Unprocessed deposits that arrived after expiry / cancellation (daily, 1.1.0)
`webhook.signing_secret_missing`	A merchant webhook that could not be sent for lack of a signing secret (once per 24 hours per merchant × environment, 1.1.0)
`blacklist.candidate`	Repeated-failure detection

11.5 Not implemented / planned (chapter 11)

- Merchants cannot edit the wording of the mail sent to them (templates are edited by the PSP).
- No automatic statement mail and no webhook / e-mail delivery of announcements.
- A real mail provider must be added to `lib/email/send.ts`.

---

12. Testing and the sandbox

12.1 Test card numbers and scenarios

In the test environment (`wp_test_ak_` key), sending one of these numbers in `card.number` reproduces a scenario (`lib/public-api/test-cards.ts`). Any other Luhn-valid number succeeds; Luhn failures answer 400 `invalid_input`. In live (`wp_live_ak_`) these numbers answer 400 `test_card_in_live`.

Number	Outcome (**`outcome`**)	Transaction **`status`**	**`failureReason`**	Description
`4242424242424242`	`success`	`captured` (`authorized` with `autoCapture: false`)	—	Success (authorized and captured). Visa
`5555555555554444`	`success`	Same	—	Success (Mastercard)
`3566002020360505`	`success`	Same	—	Success (JCB)
`4000000000003220`	`authorized_only`	`authorized`	—	3-D Secure style: stops at authorized; complete from the PSP dashboard
`4000000000000002`	`declined`	`failed`	`card_declined`	Declined by the issuer
`4000000000009995`	`insufficient_funds`	`failed`	`insufficient_funds`	Insufficient funds
`4000000000000069`	`expired_card`	`failed`	`expired_card`	Expired card
`4000000000000127`	`incorrect_cvc`	`failed`	`incorrect_cvc`	Incorrect CVC
`4000000000000119`	`processing_error`	`failed`	`processing_error`	Processing error at the network

- Any `expMonth` 1–12 and `expYear` 2024–2099 is accepted (no past-date check). `cvc` is neither stored nor validated.

- Failure scenarios return HTTP 201 with `status: "failed"` and appear in the portal. No `payment.failed` webhook is sent (creation-time failure). They do feed repeated-failure detection (by default 3 failures in 10 minutes from the same card / IP / e-mail create a blacklist candidate).
- `credit` / `qr` transactions without `card` succeed without card details.

12.2 Testing konbini and bank transfers

Method	How to complete in test
Konbini	(a) PSP dashboard transaction detail "Reconcile deposit", (b) demo notification API `POST /api/demo/cvs-notify`, (c) the payment page's "Mark as paid at the store (demo)"
Bank transfer	(a) PSP dashboard "Reconcile deposit" (short / excess reproducible), (b) demo notification API `POST /api/demo/deposit-notify`
Expiry	Wait for the expiry cron (`/api/cron/expire-transfers`), or the PSP dashboard's "Expire" (only after the deadline)
Late deposit / environment mismatch	Calling the demo deposit API with an expired transfer ID, or with a `mode` that differs from the transaction's environment, answers 202 and holds the deposit as unmatched. On the PSP dashboard's "Unmatched deposits" it can be applied as a late deposit (reason required) or dismissed
Re-sent notification	Sending the same `eventId` twice answers 200 `{ duplicate: true }` the second time and applies nothing

12.3 Demo endpoints

The demo deposit APIs are protected by the PSP-side environment variable **`DEPOSIT_NOTIFY_SECRET`**: unset in production → 404, unset in development → open. Merchants that want to use them need the secret from the PSP (the receiver-interface specification is in Specification B, chapter 3).

Endpoint	Request	Response
`POST /api/demo/deposit-notify`	`Authorization: Bearer <DEPOSIT_NOTIFY_SECRET>`, `{ eventId?, transferId, amount, mode, payerName?, bankReference?, receivedAt? }`. **`mode`** (**`test`** / **`live`**) is mandatory (missing → 400 `invalid_input`, `fieldErrors.mode = "required"`); `eventId` is an optional de-duplication key of ≤128 characters (1.1.0)	Exact / excess: 200 `{ matched: true, notificationId, notificationNo, duplicate: false, status: "captured", depositDiff: null | "excess", completed: true, … }`; short: 200 `{ status: "awaiting_transfer", depositDiff: "short", completed: false }`; unknown transfer ID / environment mismatch / expired / cancelled: 202 `{ matched: false, reason: "not_found" | "mode_mismatch" | "late_deposit" | "cancelled_deposit", unmatchedDepositId, paymentId }` (held as an unmatched deposit); a resend (same `eventId`, or identical content): 200 `{ duplicate: true, notificationId, paymentId }` without re-applying; completed etc.: 409 `invalid_transition`
`POST /api/demo/cvs-notify`	Same Bearer, `{ eventId?, receiptNo (13 digits, hyphens allowed), amount?, receivedAt?, mode }` (`mode` mandatory)	200 `{ receiptNo, paymentId, status, depositDiff, completed, notificationId, … }`; environment mismatch / expired / cancelled: 202 (as above); unknown: 404 `receipt_not_found`; resend: 200 `{ duplicate: true }`
`GET / POST /api/demo/webhook-receiver?result=ok|error|timeout`	None	Simulated webhook receiver: `ok` = 200 (echoes the request), `error` = 500, `timeout` = waits 12 seconds. The seeded merchants' notification URL points here

Every deposit notification (applied / duplicate / unmatched / rejected) is recorded in the PSP dashboard's "Deposit notification log" (`/deposit-notifications` , number `DN-00001`) and linked from the transaction detail's "Related deposit notifications / unmatched deposits" (1.1.0). The full contract is in Specification B, chapter 3.

12.4 Using the sandbox console

1. Log in to the portal and find your test key (`wp_test_ak_…`) under "API keys" (ask the PSP if none exists).
2. Open "Sandbox", paste the key and pick an endpoint; "Load example" fills the body.
3. "Send" calls the real **`/api/v1`** from the browser and shows the HTTP status, duration, JSON response and an equivalent curl command.
4. The transactions you create appear in the portal's "Transactions" and on the PSP dashboard, and webhooks are really delivered.

Constraints: live keys are refused, the key lives only in the browser's sessionStorage, and the 60 / minute rate limit applies.

12.5 Testing webhook reception

1. Have the PSP register your https notification URL and give you the test signing secret.
2. Create a `credit` transaction with `autoCapture: true` (sandbox or API): `payment.completed` arrives immediately.
3. Request a refund with `POST /api/v1/refunds` and ask the PSP to approve it: `refund.completed` arrives.
4. Return 5xx or delay the response to observe the 1 min → 5 min → … retries and the records in the portal's "Webhook log".
5. Resend manually and confirm that `X-Timestamp` and the signature are recomputed.

12.6 Test data (seed)

The demo environment ships with these merchants and keys (`prisma/seed-data.ts` ; reset daily at 03:00 JST).

Item	Value
Merchants	`MER-00001` (ハンドメイド雑貨 コトリ) and 3 others
Test API key (MER-00001)	`wp_test_ak_KotoriDemoTestKey000000000000001`
Merchant login	Shown on the login page's demo accounts panel (when `DEMO_LOGIN_HINT` is set)
Notification URL	`/api/demo/webhook-receiver`

12.7 Not implemented / planned (chapter 12)

- No scenario numbers (test-card equivalents) for QR, konbini or bank transfer.
- No manual webhook firing of arbitrary events in test (the PSP dashboard's "Simulate webhook" tests the provider → Wire Pay direction).

13. Go-live checklist

Items a merchant (or downstream PSP / wallet) confirms before going live. Section numbers in brackets point to the details.

13.1 Contract and account

- ☐ Received the merchant ID (`MER-xxxxx`) and password from the PSP and can log in at `/merchant/login` (3.3)
- ☐ The methods you will use are enabled in the contract ("Available" on `/merchant/methods`) (4.11)
- ☐ Agreed the transfer ID length (7 / 8) and the transfer, konbini and refund-window day counts with the PSP (2.4, 8.1)
- ☐ Agreed the limits (per transaction / daily / monthly) (10.2)
- ☐ The PSP registered the settlement settings (cycle, payout term, bank account, settlement fee, registration number and address) (9.1)
- ☐ Reviewed the contract plan and fee rates (9.4)

13.2 API keys

- ☐ Received and safely stored the test API key (`wp_test_ak_`) and test signing secret (`wp_test_ss_`) (3.2)
- ☐ `GET /api/merchant-api/ping` or `GET /api/v1/transactions?limit=1` authenticates with the test key (5.8, 5.4)
- ☐ Received the live API key (`wp_live_ak_`) and live signing secret (`wp_live_ss_`) (at go-live)
- ☐ Rotation procedures are part of operations (test: self-service; live: ask the PSP; 7-day grace) (3.2)
- ☐ Told the PSP your API servers' egress IPs and confirmed the merchant-API IP restriction (10.1)

13.3 Creating transactions

- ☐ `POST /api/v1/transactions` sends an `Idempotency-Key` and retries with the same key on 5xx / network failure (5.1)
- ☐ The code branches on the response `status` and explains `failureReason` to the user when `failed` (5.3)
- ☐ `bank`: `bankTransfer.transferId` and `dueAt` are shown to the customer with "prefix the transfer ID to the payer name" (7.3)
- ☐ `convenience`: `konbini.receiptNo` (shown 4-4-5), `brand` and `dueAt` are shown to the customer, and `status` `awaiting_payment` is treated as "awaiting payment" (7.4)
- ☐ `credit` / `qr`: the `autoCapture` policy is decided; with authorization only, the PSP completion procedure is agreed (7.2)
- ☐ 429 `rate_limited` is retried after `Retry-After` (5.1)
- ☐ The 9 test cards were exercised: success, authorized only, each failure (12.1)

- [ ] `customerId` is a stable customer identifier of your own (customer master key) (10.6)

13.4 Receiving webhooks

- [ ] An https notification URL is registered with the PSP and the format (JSON / form / query), parameter set and identifier format are chosen (6.2)
- [ ] `X-Timestamp` + `X-Signature` are verified with HMAC-SHA256 over the raw body (or the GET query string) in constant time (6.4)
- [ ] Notifications whose timestamp is off by 5 minutes or more are rejected (6.4)
- [ ] Both the test and the live signing secret can verify (6.4)
- [ ] The portal's "API keys" shows a usable signing secret for each environment (no webhook is sent to an environment without one) (6.4)
- [ ] Processing is idempotent on `payment_id` + `event` and order-independent (6.7)
- [ ] 2xx is returned within 10 seconds (heavy work asynchronous) (6.6)
- [ ] Every event (`payment.completed` / `payment.failed` / `payment.expired` / `payment.cancelled` / `refund.completed` / `chargeback.opened` / `limit.exceeded` / `subscription.*`) gets a 2xx, and unknown events are ignored safely (6.1)
- [ ] Successes are visible in the portal's "Webhook log" and the resend procedure is part of operations (4.7, 6.8)

13.5 Refunds, settlement and others

- [ ] A refund request (API or portal) and the `refund.completed` webhook were verified (8.1)
- [ ] The procedure for refunds past the window (ask the PSP) is in place (8.1)
- [ ] Settlement batches / lines (portal, `/api/merchant-api/settlements`) and the invoice PDF were checked (9.7, 9.8)
- [ ] A way to read announcements / incidents (portal or `GET /api/merchant-api/announcements`) is chosen (11.3)
- [ ] Customer mail texts (transfer, konbini, completion, refund) were reviewed with the PSP (11.2)
- [ ] Agreed the merchant e-mail notification setting (per payment / daily digest / none) and its recipient (contact e-mail) with the PSP (11.1)
- [ ] The personal-data retention (default 24 months) and the effect of anonymisation are understood (10.5)

13.6 Switching to live

- [ ] Switched to the live key and no test card numbers are sent (no `test_card_in_live`) (12.1)
- [ ] Transactions, webhooks and settlements were checked with the header switch set to "Live" (1.3)
- [ ] Webhook verification works with the live signing secret (6.4)
- [ ] If the allow-list mode is on, every egress IP of the live API servers is registered as an allow rule (10.1)

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14. Not implemented / planned

A consolidated list of the items scattered at the chapter ends. None of them exists at the target commit and none is described in the body.

14.1 Payment processing

Item	Status
Live connection to external payment networks	None. Card / QR complete immediately in the DB; konbini / bank deposits are simulated by the demo APIs or PSP actions (Specification B, chapter 11)
Real card entry, 3-D Secure or QR display on the payment page	None ("Complete payment (demo)")
Re-charging a stored card	None
Test scenario numbers for QR, konbini, bank transfer	None

14.2 Merchant API

Item	Status
Capture / cancel / expire API	None (PSP dashboard actions)
Reading, listing or cancelling refund requests	None
Reading, listing or disabling payment links	None (disable on the PSP dashboard)
Customer, stored-card and subscription APIs	None
Retrieving / rotating signing secrets by API	None
Webhook event subscriptions and an environment field	None
Validation of `GET /transactions` `status`	None (unknown values return zero rows)

14.3 Merchant portal

Item	Status
Password change / reissue, multiple staff, two-factor	None
Issuing / rotating / revoking live keys	None (ask the PSP)
Self-service changes to notification settings, enabled methods, deadlines	None
CSV export, detailed transaction filters, webhook delivery detail (headers / body)	None
Viewing IP restrictions, limits, blacklist	None
Chargeback viewing and evidence upload	None

14.4 Notifications and settlement

Item	Status
Real mail delivery	`EMAIL_PROVIDER=log` records only
Automatic settlement generation, payout API, automatic statement mail	None
Webhooks for refund rejection and chargeback won / lost / confirmed	None
Webhook / e-mail delivery of announcements	None

14.5 Security

Item	Status
Column-level encryption of personal data	None (plain columns + masking + retention)
IPv6 IP restrictions	Not evaluated
Fixed webhook egress IP	None (verify by signature)

Items removed from this list because 1.1.0 implemented them: merchant mail driven by the e-mail notification setting (11.1), receipts issued from the portal (4.5), the merchant API rate limit (5.8), de-duplication of deposit notifications and holding of late deposits (7.3), fail-closed webhook signing (6.4), enforcement of IP allow rules (10.1).

Appendix A: Error codes

Error codes a merchant may receive. Response format: 5.1.

A.1 Public API (`/api/v1/*`) — identical to `x-error-codes`

code	HTTP	Meaning
`invalid_api_key`	401	Missing, malformed, revoked or expired key, or a signing secret used for authentication
`ip_blocked`	403	Refused by the merchant's IP restrictions (a block rule, or no matching allow rule while the allow-list mode is on; `details.reason` = `deny` / `not_in_allowlist`)
`invalid_input`	400	Validation failed; `details.fieldErrors` names the fields
`invalid_json`	400	Body is not JSON
`test_card_in_live`	400	Test card number in live
`method_not_enabled`	400	Method not in the merchant's contract

code	HTTP	Meaning
`provider_not_found`	400	No active provider in the environment
`method_suspended`	409	Payment method suspended (`details.scope` , `details.method`)
`not_found`	404	Missing, or another merchant's / environment's resource
`refund_exceeds`	400	Refund exceeds the refundable balance
`refund_window_expired`	409	Past the refund window
`invalid_transition`	409	Operation not allowed in the current status (e.g. refund of an incomplete transaction)
`demo_limit`	403	Demo data cap
`rate_limited`	429	Over 60 / minute; see `Retry-After`
`internal`	500	Unexpected server error

The shared handler may additionally return `conflict` (409, uniqueness violation), `provider_disabled` (400), `encryption_key` (500, PSP-side key misconfiguration) and `undecryptable` (422).

A.2 Merchant portal APIs (**`/api/merchant-portal/*`**)

code	HTTP	Where
`invalid_credentials`	401	Login: wrong merchant ID or password
`merchant_locked`	423	Login: locked after 5 consecutive failures (`{minutes}`)
`unauthorized`	401	No / expired merchant session
`csrf_origin` / `csrf_header`	403	Missing same-origin / `X-Requested-With: fetch`
`rate_limited`	429	Per-IP cap (mutations 60 / min, login 10 / min)
`not_found`	404	Another merchant's transaction / delivery / key / batch
`invalid_transition` / `refund_exceeds` / `refund_window_expired` / `invalid_input`	409 / 400 / 409 / 400	Refund request (as in A.1)
`live_key_rotation`	403	Rotating a live key
`key_not_active`	409	Rotating a revoked / expired key
`settlement_not_confirmed`	409	Invoice PDF of a collecting batch
`invoice_no_missing`	409	Batch without an invoice number (batches confirmed before 1.1.0 are numbered by `npm run db:backfill`)

code	HTTP	Where
`receipt_not_completed`	409	Receipt for a transaction that is not completed / partially refunded (portal receipt issue)

A.3 Hosted payment page APIs (`/api/pay/*`)

code	HTTP	Meaning
`not_found`	404	Unknown token, or a transaction not belonging to the link
`link_disabled` / `link_expired` / `link_exhausted`	409	Disabled / expired / usage cap reached
`invalid_input`	400	Invalid input (including `fieldErrors.method = unavailable`)
`invalid_transition`	409	"Complete" on an already completed transaction
`receipt_link_invalid`	403	Receipt URL signature mismatch or expired
`receipt_not_completed`	409	Receipt for an incomplete transaction

A.4 Demo deposit APIs (`/api/demo/*`)

code	HTTP	Meaning
`unauthorized`	401	`DEPOSIT_NOTIFY_SECRET` mismatch
`not_found`	404	Secret unset in production (disabled)
`invalid_input`	400	Missing `mode` (`fieldErrors.mode = "required"`), invalid `amount` or `receivedAt`, etc.
`receipt_not_found`	404	Unknown receipt number (konbini)
`invalid_transition`	409	Deposit on a transaction that is completed or otherwise not awaiting transfer / payment. Deposits on expired / cancelled transactions answer 202 and are held as unmatched instead (1.1.0)
`not_bank_transfer`	400	Transaction with neither transfer ID nor receipt number

Appendix B: Statuses

Object	Values (label ja / en)
Transaction (`Payment.status`)	The 11 values of 2.2 (including `awaiting_payment`)
Refund request (`Refund.status`)	`requested` 申請中 / Requested, `approved` 承認 / Approved, `rejected` 否認 / Rejected, `processing` 処理中 / Processing, `completed` 完了 / Completed, `failed` 失敗 / Failed
Refund kind (`Refund.kind`)	`full`, `partial`

Object	Values (label ja / en)
Refund reason (`reasonCode`)	`customer` customer request, `defect` defective goods, `duplicate` duplicate payment, `other`
Chargeback (`Chargeback.status`)	`received` , `disputed` , `won` , `lost` , `confirmed`
Chargeback reason	`fraud` , `not_received` , `not_as_described` , `duplicate` , `other`
Settlement batch (`SettlementBatch.status`)	`collecting` , `confirmed` , `scheduled` , `paid` , `held`
Settlement line kind (`SettlementLine.kind`)	`payment` sale, `refund` , `chargeback`
Webhook delivery (`WebhookDelivery.status`)	`pending` , `success` , `retrying` , `failed` , `abandoned`
Subscription (`Subscription.status`)	`trialing` , `active` , `past_due` , `suspended` , `cancelled`
Payment link (`PaymentLink.status`)	`active` , `disabled`
API key	`active` , `expiring` , `expired` , `revoked`
Short / excess deposit (`Payment.depositDiff` / `depositDiffStatus`)	`short` / `excess` ; follow-up `open` , `refund_arranged` , `notified` , `resolved`
Announcement (`Announcement.status` / `kind`)	`draft` , `published` , `ended` ; kinds `notice` / `maintenance` / `incident` ; incident updates `occurred` / `investigating` / `resolved`
Customer	`anonymizedAt` set = anonymised
Deposit notification (`DepositNotification.result`)	`applied` , `duplicate` , `unmatched` , `rejected` ; `resultReason` : `mode_mismatch` / `late_deposit` / `cancelled_deposit` / `not_found` / `no_id` / `invalid_status` / `apply_failed`
Unmatched deposit (`UnmatchedDeposit.status` / `reason` / `dismissKind`)	`unmatched` , `matched` , `dismissed` ; reasons `late_deposit` / `cancelled_deposit` / `mode_mismatch` ; dismissal kinds `returned` (return arranged) / `other`
Receipt origin (`Receipt.source`)	`dashboard` PSP dashboard, `customer` payment page, `merchant` merchant portal

Appendix C: Webhook events

Section 6.1 is authoritative. Event names are `MERCHANT_WEBHOOK_EVENTS` in `lib/webhooks/events.ts` (10): `payment.completed` , `payment.failed` , `payment.expired` , `payment.cancelled` , `refund.completed` , `chargeback.opened` , `limit.exceeded` , `subscription.charged` , `subscription.payment_failed` , `subscription.suspended` .

Appendix D: Screens

D.1 Merchant portal and customer-facing

Path	Screen	Section
`/merchant/login`	Merchant login	4.2
`/merchant`	Home	4.3

Path	Screen	Section
`/merchant/transactions`	Transactions	4.4
`/merchant/transactions/[id]`	Transaction detail (refund request)	4.5
`/merchant/refunds`	Refund requests	4.6
`/merchant/webhooks`	Webhook log (resend)	4.7
`/merchant/settlements`	Settlements	4.8
`/merchant/settlements/[id]`	Settlement detail (invoice PDF)	4.9
`/merchant/api-keys`	API keys (test-key rotation)	4.10
`/merchant/methods`	Payment methods	4.11
`/merchant/announcements`	Announcements	4.12
`/merchant/customers`	Customers	4.13
`/merchant/customers/[id]`	Customer detail	4.14
`/merchant/sandbox`	Sandbox	4.15
`/pay/[token]`	Hosted payment page	4.16
`/docs/api`, `/docs/api/[chapter]`	Public API docs (Getting started / Authentication & environments / Errors & rate limits / Test cards / Webhooks / Sandbox + generated reference)	—

D.2 PSP dashboard screens relevant to merchants (reference)

Path	Screen
`/merchants`, `/merchants/[id]?tab=basic|contract|limits|apikeys|notify|settlement|webhooks|methods`	Merchants (detail tabs)
`/applications`, `/payments`, `/payments/[id]`	Payment applications / Payments / Transaction detail (complete, fail, cancel, reconcile, refund request, API re-notify, receipt)
`/refunds`, `/refunds/[id]`	Refunds (approve / reject)
`/settlements`, `/settlements/[id]`	Settlements
`/webhook-logs`, `/webhook-logs/[id]`	Webhook log (detail, resend)
`/chargebacks`, `/chargebacks/new`, `/chargebacks/[id]`	Chargebacks
`/unmatched-deposits`, `/deposit-diffs`, `/deposit-notifications`	Unmatched deposits (match, late deposit, return) / Short & excess deposits / Deposit notification log
`/payment-links`, `/payment-links/[id]`	Payment links (disable, QR)
`/subscriptions`, `/subscriptions/[id]`	Subscriptions
`/payment-method-controls`	Payment method suspension (global)
`/ip-restrictions`, `/blacklist`	IP restrictions (rules, allow-list mode) / Blacklist
`/announcements`	Announcements
`/customers`, `/customers/[id]`	Customers (masked, anonymisation)

Path	Screen
`/email-templates`, `/email-logs`	E-mail templates / E-mail log

Appendix E: Permission matrix (complete)

Permissions per PSP dashboard role (`admin` / `accounting` / `cs` / `developer` / `viewer`). The single source is **`PERMISSIONS`** in **`lib/role.ts`**; the table below is generated by `npx tsx scripts/gen-permissions.ts` (never edited by hand). The rows that concern merchants directly are excerpted in 3.5. Anonymous visitors can only read; with `DEMO_READONLY=1` deletions and provider updates answer 403 for everyone.

Feature	Action	admin	accounting	cs	developer	viewer
Merchants	View	○	○	○	○	○
Merchants	Create	○		○		
Merchants	Update	○		○		
Merchants	Reveal password	○				
Merchants	Regenerate password	○				
Contract & fees / limits / deadlines / settlement settings	View	○	○	○	○	○
Contract & fees / limits / deadlines / settlement settings	Update	○	○			
API keys and signing secrets	View	○	○	○	○	○
API keys and signing secrets	Issue	○			○	
API keys and signing secrets	Rotate	○			○	
API keys and signing secrets	Revoke	○			○	
Agents	View	○	○	○	○	○
Agents	Create	○		○		
Agents	Update	○		○		
Transactions	View	○	○	○	○	○
Transactions	Create	○		○		
Transactions	Complete / fail / cancel / expire	○				
Transactions	Reconcile deposit	○	○			
Transactions	API re-notify	○		○	○	
Transactions	Simulate webhook	○			○	
Transaction notes	View	○	○	○	○	○
Transaction notes	Create	○	○	○	○	
Transaction notes	Edit own notes	○	○	○	○	
Transaction notes	Delete	○				

Feature	Action	admin	accounting	cs	developer	viewer
Transaction notes	Pin	○		○		
Unmatched deposits	View	○	○	○	○	○
Unmatched deposits	Register	○	○	○		
Unmatched deposits	Match	○	○			
Unmatched deposits	Dismiss	○	○	○		
Deposit notification log	View	○	○	○	○	○
Deposit differences	View	○	○	○	○	○
Deposit differences	Update	○	○	○		
Refunds	View	○	○	○	○	○
Refunds	Request	○		○		
Refunds	Force past the window	○				
Refunds	Approve / reject	○	○			
Receipts	View	○	○	○	○	○
Receipts	Issue	○	○	○		
Settlement batches	View	○	○	○	○	○
Settlement batches	Generate	○	○			
Settlement batches	Confirm / schedule / pay / hold / release	○	○			
Settlement batches	Delete	○	○			
Invoices	Export	○	○			
Agent fees	View	○	○	○	○	○
Reports	View	○	○	○	○	○
Webhook deliveries	View	○	○	○	○	○
Webhook deliveries	Resend	○			○	
Chargebacks	View	○	○	○	○	○
Chargebacks	Create	○	○			
Chargebacks	Update	○	○			
Blacklist	View	○	○	○	○	○
Blacklist	Create	○		○		
Blacklist	Update	○		○		
Detection rule	View	○	○	○	○	○
Detection rule	Update	○		○		
E-mail templates	View	○	○	○	○	○
E-mail templates	Update	○			○	

Feature	Action	admin	accounting	cs	developer	viewer
E-mail log	View	○	○	○	○	○
Announcements	View	○	○	○	○	○
Announcements	Create	○		○		
Announcements	Update	○		○		
Announcements	Publish / end / add update	○		○		
Billing plans	View	○	○	○	○	○
Billing plans	Create	○		○		
Billing plans	Update	○		○		
Subscriptions	View	○	○	○	○	○
Subscriptions	Create	○		○		
Subscriptions	Charge / pause / resume / cancel	○		○		
Demo charge failure rate	View	○	○	○	○	○
Demo charge failure rate	Update	○			○	
Payment links	View	○	○	○	○	○
Payment links	Create	○		○		
Payment links	Disable	○		○		
IP restrictions	View	○	○	○	○	○
IP restrictions	Create	○			○	
IP restrictions	Update	○			○	
IP restrictions	Delete	○			○	
Payment providers	View	○	○	○	○	○
Payment providers	Create	○			○	
Payment providers	Update	○			○	
Payment providers	Delete	○			○	
Payment providers	Connection test	○			○	
Payment method suspension	View	○	○	○	○	○
Payment method suspension	Suspend globally	○				
Payment method suspension	Suspend per merchant	○		○		
Admin accounts	View	○				
Admin accounts	Create	○				
Admin accounts	Update	○				
Admin accounts	Temporary password / 2FA reset	○				
Own account	Update	○	○	○	○	○
Audit log	View	○	○	○	○	○

Feature	Action	admin	accounting	cs	developer	viewer
Global search (admins)	View	○				
Global settings	View	○				
Global settings	Update	○				
Customers	View	○	○	○	○	○
Customers	Anonymise	○				
CSV export	Export	○	○	○	○	
Personal data	Unmask	○		○		
API docs	View	○	○	○	○	○

Appendix F: Frequent questions → where to look

Question	Check first	Sections
A deposit is not reflected (bank transfer / konbini)	The transaction's status and deadline (`expired` means the deposit is held as a late deposit among the unmatched deposits). Whether the transfer ID was written correctly. Whether the notification's `mode` matches the transaction's environment. Whether the PSP dashboard's "Deposit notification log" / "Unmatched deposits" show the notification	2.4, 7.3, 7.4, 12.3, Specification B 3.2–3.5 / 5.3
Webhooks do not arrive	Status and response in the portal's "Webhook log" (`failed` / `abandoned` ; `error: signing_secret_missing` means no signing secret is issued). Whether the notification URL is https and answers 2xx within 10 seconds. Whether you verify with the secret of the right environment (test / live). Resend from the log	4.7, 6.4, 6.6, 6.7, 6.8
A refund is refused	Whether the transaction is `captured` / `partially_refunded` (`authorized` is cancelled, `chargeback` cannot be refunded). The refundable balance (amount − refunded − pending requests) and the refund window (`refundWindowDays`). Past the window only the PSP's admin can force it	5.6, 8.1, 8.2
The settlement amount looks wrong	The line rules (sales by `capturedAt` , refunds by `completedAt` , chargebacks by `resolvedAt` within the period), the contract rate in force at completion, the settlement fee, and separate batches per environment (test / live). Invoice tax is tax-inclusive, per rate, with the configured rounding	9.2–9.5, 9.8
The live key answers 400	`test_card_in_live` (test card number in live), `method_not_enabled` (method not in the contract), `provider_not_found` (no active live provider), `invalid_input` (`details.fieldErrors`). 403 `ip_blocked` is an IP restriction (block rule / allow-list mode)	5.3, 10.1, 12.1, Appendix A
Transactions are created as `failed`	`failureReason` (limits `limit_exceeded:*` , blacklist `blocked:*` , test-card failure scenarios). The HTTP status is 201	5.3, 10.2, 10.4, 12.1
No e-mail arrives	The demo only logs mail. Customer mail needs the transaction's `customerEmail` ; merchant mail follows `emailNotification` and goes to the contact e-mail	11.1, 11.2

Revision history

Version	Date	Commit	Changes
1.0.0	2026-09-06	`6b72a40`	First edition
1.1.0	2026-09-07	`cf1c153`	Reflects the implementation changes of fix phases 1–5. Breaking: the transaction status `awaiting_payment` (2.2, 2.3, 7.4); `mode` mandatory on deposit notifications with environment matching, and 202 holding of late / cancelled deposits (7.3, 12.3, Appendix A.4); the invoice number format `INV-YYYYMM-<merchant number>-<2-digit sequence>` (9.8). Added: idempotent deposit notifications (`eventId`) and the deposit notification log, the `late_deposit` transition, the automatic test signing secret at registration (3.1), fail-closed webhook signing (6.4), merchant e-mail notification settings (11.1), the completion e-mail for immediately captured transactions (11.2), the merchant API rate limit (5.8), receipts from the portal (4.5), the IP allow-list mode (10.1), live-mode indication (4.1), the demo-only PSP link and login autocomplete (4.2), the English label `pending` = "Pending", Appendix E (permission matrix) and Appendix F (frequent questions). Details in `docs/spec/CHANGELOG.md`